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Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis

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Conflicts of Interest

The authors declare that there is no conflict of interest

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Abbreviations

AP: Acute pancreatitis

APD: Abdominal paracentesis drainage

PCD: Percutaneous catheter drainage

MAP: Mild acute pancreatitis

MSAP: Moderate severe acute pancreatitis

SAP: Severe acute pancreatitis

RIS: Required information size

CLS: Capillary leak syndrome

IAP: Intra-abdominal pressure

RCT: Randomized controlled trial

OR: Odds ratio

CI: Confidence interval

SMD: Standardized mean difference

CNKI: China National Knowledge Infrastructure

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Abstract

Objectives

This systematic review and meta analysis was performed to investigate whether abdominal paracentesis drainage (APD) is more advantageous in treating acute pancreatitis (AP) patients when comparing with conventional "step-up" method, and evaluate the risk of infectious complications caused by APD.

Design

Systematic review and meta-analysis.

Methods

PubMed, EMBASE, Cochrane library, OVID, CNKI and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) from inception to July 25, 2020. Studies related to comparing APD and conventional "step-up" method for curing AP patients were included.

Outcomes

Primary outcome was all-cause mortality. The secondary outcomes were the rate of organ failure, infectious complications, hospitalization expenses and length of hospital stay.

Results

A total of five cohort studies and three RCTs were included. Compared with conventional "step-up" method, pooled results suggested that APD significantly decreased all-cause mortality (OR 0.45, 95%CI 0.25-0.83, *P*=0.01), length of hospital stay (Cohort studies: SMD -0.32, 95%CI -0.51, -0.12, *P*=0.002; RCTs: SMD -0.45, 95%CI -0.64, -0.26, *P*<0.001) and hospitalization expenses (Cohort studies: SMD -2.49, 95%CI -4.46, -0.51, *P*<0.001; RCTs: SMD -0.67, 95%CI -0.89, -0.44, *P*<0.001). In addition, no evidence proved that APD increased the incidence of infectious complications. However, there were some difference between cohort studies and RCTs subgroup in the incidence of organ failure (Cohort studies: OR 0.65, 95%CI 0.34-1.26, *P*=0.2; RCTs: OR 0.58, 95%CI 0.38-0.89, *P*=0.01).

Conclusions

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Current evidence showed that APD could significantly decreased all-cause mortality, expenses and the le ngth of stay during hospitalization, while extra infectious risk was not increased. The conclusions are n eeded to be verified indeed by more high-quality RCTs due to the limited quality of included trials.

KEY WORDS: abdominal paracentesis drainage; acute pancreatitis; percutaneous catheter drainage; syste matic review; meta-analysis

Strengths and limitations of this study

1. This trial was the first systematic review and meta analysis comparing APD with conventional "step -up" method in treating AP patients.
2. The results favored the positive effect of APD in terms of reducing all-cause mortality, expenses, th e length of stay during hospitalization, hence it provides support for the further clinical application of APD.
3. This study relied greatly on cohort studies due to limited number of published RCTs, and despite s ensitivity analysis proved the stability of results of spending, the heterogeneity of it was very large.
4. This meta-analysis involved a relatively small number of studies, and the small study effect also sh ould be cautious.

1. Introduction

Acute pancreatitis (AP), characterized by rapid progression, severe complications and high mortality, is one of the most common diseases causing admission to the intensive care units [1]. AP was divided into acute pancreatitis (MAP), moderate severe acute pancreatitis (MSAP) and severe acute pancreatitis (SAP) according to the updated 2012 Atlanta Classification, MAP only have upper abdominal pain, nevertheless local or systemic complications, such as peripancreatic collection, will occur in MSAP, SAP stage, and persistent organ failure eventually [2].

Currently, the treatment of AP has undergone a winding exploration for a long history. In the past, conservative treatments including active rehydration and high-dose antibiotics have been used in early stage, necrosectomy was performed when previous treatment would not effectively prevent deterioration [3]. However, open surgery

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was associated with high mortality, infectious complications and prolonged recovery time[4]. In 1998, Freeny et al firstly reported a novel treatment, which commonly known as percutaneous catheter drainage (PCD), then the efficacy and safety of which were confirmed by many clinical trials [5]. Since then, the AP treatment period has entered a PCD-based "step-up" time [6,7], the purpose of which was aimed at controlling sources of infection progressively and reasonably instead of removing necrosis immediately, thereby reducing the risk of increased death caused by traumatic surgery[8]. However, this conventional PCD-based "step-up" treatment could not meet the earlier clinical requirements, because PCD must be performed at least 4 weeks after onset, when the collection of walled-off necrosis was formed [9]. However, in the early stage of AP, the release of a large number of inflammatory mediators leads to an increase in tissue capillary permeability, in hence, the fluid generated and dissiciated into the abdominal cavity or even the pelvic cavity. These toxic substances destructed the intestinal mucosal barrier and then be absorbed into the blood through the peritoneum, causing inflammatory cascades, in hence, aggravating systemic inflammatory response syndrome (SIRS). Recently, a retrospective cohort study reported a new insert catheters technique, called abdominal paracentesis drainage (APD), which was carried out through right paracolic sulci or left paracolic sulci, when the abdominal collection volume was greater than 50ml [10]. The results of this study showed that early APD treatment could effectively reduce the release of inflammatory factors and improve the clinical prognosis. However, the controversy has mainly focused on whether the treatment process would bring exogenous infection or not.

Therefore, we systematically reviewed available articles to investigate whether the APD could significantly decrease the mortality when compared with conventional "step-up" method in treating AP patients. The secondary objectives were to explored the risk of infectious complications, hospitalization expenses and length of hospital stay.

2. Material and Methods

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The present systematic review and meta-analysis was reported in accordance to the Preferred Reporting item for Systematic Review and Meta-analysis (PRISMA) statement [11]. And, this review has been registered on the PROSPERO website as No. CRD42020168537.

2.1 Data sources and searches

PubMed, EMBASE, Cochrane library, OVID, China National Knowledge Infrastructure (CNKI) and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) about the abdominal paracentesis drainage for curing AP patients from inception to February 15, 2020 by using keywords combined with Medical Subject Headings (MeSH) terms, regardless of language and region. We retrieved it again on July 25, 2020 to included newly published paper. In addition, the reference lists of relate literature were manually searched for possible trials. The search strategy for PubMed was showed in Supplementary Table 1.

2.2 Study selection

Zongqing Lu and Xingxing Zhu carried out search independently, according to predefined inclusion and exclusion criteria. Duplicate literatures deletion, title and abstract screening for relevance had been done using Endnote software. Then, the full-text was acquired to determine inclusion eligibility eventually. Any disagreement was resolved through discussion, a third review author (Min Yang) participated in where necessary.

Published RCTs and cohort studies meeting the following criteria were included: 1.population: Adult patients with acute pancreatitis regardless pathogenesis, however, detailed diagnostic criteria for acute pancreatitis must be required, and each participant must conform to requirement of APD: 1) enough volume of celiac fluid collections (> 50mL); 2) a feasible pathway existed. No limitation on race and nationality. 2.intervention: APD was administered within 4 weeks of onset. 3.comparison: traditional “step-up strategy” in which APD was not

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performed. 4.the sample size of each group exceeds 20.

2.3 Outcomes and Data Extraction

Two authors (Zongqing Lu and Di Jia) independently extracted data using a pre-piloted form designed by Excel 2019 software (Microsoft Corporation), and the results confirmed by another author (Tianfeng Hua). The collected data include: first author, publish year, sample size, mean age and sex ratio of each group, study period, cause of AP, APACHE Ⅱ scores at admission, catheter diameter and outcomes data. If any information above was inadequate, we contacted the original author via email to consult related data. We resolved discrepancies through discussion. The predefined primary outcomes were all-cause mortality during hospitalization. The secondary outcomes were the rate of organ failure, the rate of related infectious complications, spending and the length of hospital stay.

2.4 Quality Assessment

The quality of filtered articles has been assessed by two authors (Wenyan Xiao and Jin Zhang) respectively. Cochrane Handbook for Systematic Reviews of Interventions (5.1.0) was used to assess the risk of bias for RCTs, which contain seven aspects: random sequence generation, allocation concealment, performance bias, detection bias, attrition bias, reporting bias and other bias [12]. We reviewed each RCT and divided them into high, low or unclear risk of bias. However, blinding of patients and clinicians was often unrealistic in invasive treatment, and which has limited influence in primary outcomes. So, trials with more than one high risk aspect except performance bias were considered as high risk of bias whereas trials with low risk of bias for all aspect except performance bias were considered to be at low risk of bias, otherwise it was considered as unclear risk of bias. Meanwhile cohort studies was assessed using the Newcastle-Ottawa Scale (NOS), in which there were three key domains: 1.selection of study population, with four entries, 1) representativeness of the exposed cohort, 2) selection method for non-exposed cohort, 3) methods for exposure factors determined, 4) demonstration that the

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outcome was not present at the beginning of the trial. 2.Inter-group comparability with one entry regarding comparability of cohorts in the stage of design and statistical analysis. 3.Measurement of outcomes, with three entries, 1) adequate assessment of the outcome related, 2) adequate duration of follow-up, and 3) adequate follow-up in each cohort. Total score of NOS scale is 9, scores of 0-3, 4-6, and 7-9 were respectively seem as the low, moderate, and high quality of studies [13].

2.5 Quality of evidence

Two authors assess the quality of each evidence respectively by using the GRADE system (Grading of Recommendations Assessment, Development and Evaluation) for risk of bias, inconsistency, indirectness, imprecision, publication bias, large effect, confounder and dose-response gradient [14]. The quality was divided into very low, low, moderate, or high. The results were generated by using the GRADE Profiler.

2.6 Statistical Analysis

RevMan 5.3 (Nordic Cochrane Centre) was used for analysis of the included studies. Odds ratio (OR) and the 95% confidence intervals (CIs) were calculated for dichotomous data by using Mantel-Haenszel method. For continuous data, standard mean difference (SMD) and 95%CIs were calculated. *P* values less than 0.05 were considered to be significant. Meta-analysis was performed if predefined outcomes from two or more studies. Statistical heterogeneity across trials was examined using the I^2 statistical tests as well as *P* values. Those with $P<0.1$ and I^2 greater than 50% were seem as significant heterogeneity. However, due to the limited number of component studies, the statistical test for heterogeneity may be not sensitive, thus, the choice of fixed or random-effects model depending on the combined statistical and clinical considerations, if the result statistical test shown that $P>0.1$ or $I^2<50\%$, and there were no clinical heterogeneity, the fixed-effect model were performed, otherwise random-effect model. We performed subgroup analysis according to the RCTs or cohort studies when sufficient data available, to avoid inappropriate pooled results influence the validity of evidence.

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The sensitivity analysis was carried out by leave-one-out method, in order to explore the sources of heterogeneity and examined the stability of results. We performed funnel plot and Egger's test to detect the possible publication bias.

3. Results

3.1 Literature Research

The flow diagram shown the process of literature screening, selection and reasons for exclusion (Fig.1). Our initial search yielded 299 records. After removing duplications and reviewing the titles/abstracts, 23 articles were considered to be potentially eligible for inclusion. After reading the full-text, 15 studies were excluded for the following reasons: no available data (n=5), animal experimental (n=2), case-report (n=2), intervention measures inconsistent (n=5) and possibility of data duplication (n=1). As a result, eight studies [10,15-21] were eventually included in this meta-analysis.

3.2 Trials Characteristics

The characteristics of the eligible trials had been summarized in Table 1. There were five cohort studies and three RCTs published between 2014 and 2020, which compared the clinical prognosis of AP patients who cured by APD or conventional "step-up" approach. Population sizes ranged from 82 to 255, with a total of 1086 patients. Except for the catheter diameter and the catheter numbers, there was no statistical difference of baseline between APD and control groups in each trial, according to the report.

3.3 Risk of Bias Assessment and GRADE Profile Evidence

The details about risk of bias was respectively shown in Table 2 and Table 3. For included cohort studies, the duration of follow-up was enough when hospital stay time contained, due to these predefined outcomes just reflect a short-term effect during hospitalization. As a result, the mean of NOS score was 7.2, which indicated a high quality and a low risk of bias about cohort studies. For RCTs included, one was categorized as low risk of

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bias [16], other trials were considered as unclear risk of bias because they did not report the way of allocation concealment [18,21].

GRADE evidence profiles are showed in Supplementary Table 2. Due to either unclear risk of bias or imprecision resulted from small sample size influence and relative wide confidence interval, the quality of evidence of outcomes in RCTs subgroup was low or very low, according to the GRADE framework. In addition, we also determined the GRADE evidence of cohort studies subgroup as low or very low eventually, considering the natural risk of observational trials, and small effect size except for all-cause mortality.

3.4 Primary outcome

All-cause mortality during hospitalization was reported in six trials, which included 820 patients [10,15-17,19,20]. Compared with conventional "step-up" approach, five cohort studies [10,15,17,19,20] of APD intervention reported a significantly reduced risk of all-cause mortality in acute pancreatitis patients (OR 0.45, 95%CI 0.25-0.83, P=0.01) with low heterogeneity (I^2=0.0%) (Fig.2). However, there was no significantly differences in all-cause mortality reported by another RCT (n=161, OR 0.55, 99%CI 0.13-2.37, P=0.42) [16].

3.5 Secondary outcomes:

3.5.1 The rate of organ failure: Five studies provided data on the rate of organ failure with totaling 617 patients [10,16,18,19,21]. Compared with conventional "step-up" approach, the results of cohort studies subgroup [10,19] shown that there were no significantly differences in the incidence of organ failure (n=190, OR 0.65, 95%CI 0.34-1.26, P=0.2). However, the results of RCTs subgroup [16,18,21] supported APD (n=427, OR 0.58, 95%CI 0.38-0.89, P=0.01) (Supplementary Figure 1). We considered that the inconsistencies of results between two subgroups was result from the difference in sample size, because preferred APD trend have appeared in cohort studies subgroup. The heterogeneity in cohort studies and RCTs subgroup was I^2=0% and I^2=29%, respectively.

3.5.2 The risk of infectious complications: Five studies reported the number of infectious complications, which

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include microbial infection [10,19-21], sepsis [10,15,20] and bacteremia [10,15,20], with totaling 695 patients. We pooled data from three cohort studies[10,19,20]which showed no significantly differences on the incidence of microbial infection (OR 1.05, 95%CI 0.52-2.11, *P*=0.89) with I^2=0%, and this was consistent with another RCT's result (OR 1.56, 95%CI 0.53-4.60, *P*=0.42)[21]. For the incidence of sepsis and bacteremia, the pooled results from three cohort studies showed that APD could not significantly increased the related risk versus conventional "step-up" approach (sepsis: OR 0.69, 95%CI 0.41-1.16, *P*=0.16; bacteremia: OR 0.93, 95%CI 0.58-1.48, *P*=0.75). The heterogeneity in both of them was I^2=0%. (Supplementary Figure 2)

3.5.3 Length of hospital stay: Four cohort studies (totaling 404 patients) [10,17,19,20] and three RCTs (totaling 427 patients) [16,18,21] were included in this analysis. The results of cohort studies subgroup showed that APD significantly decreased the length of hospital stay when compared with conventional "step-up" approach (SMD -0.32, 95%CI -0.51, -0.12, *P*=0.002). In addition, the pooled results from three RCTs also supported APD (SMD -0.45, 95%CI -0.64, -0.26, *P*<0.001). The heterogeneity in each subgroup was I^2=18% and I^2=0%. (Supplementary Fig.3).

3.5.4 Spending: We extracted the data about spending during hospitalization from three cohort studies and two RCTs, with totaling 601 patients [10,16,19-21]. The pooled results of cohort studies subgroup [10,19,20] demonstrated that APD could significantly reduced the spending versus conventional "step-up" approach (SMD -2.49, 95%CI -4.46, -0.51, *P*<0.001), which was consistent with the results of RCTs subgroup [16,21] (SMD -0.67, 95%CI -0.89, -0.44, *P*<0.001), and heterogeneity were I^2=97%, I^2=0% in cohort studies and RCTs subgroup respectively. (Supplementary Fig.4)

3.6 Publication Bias and Sensitivity Analysis

We just included cohort studies to explore the potential publication bias for primary outcomes, due to the limited number of RCT. The funnel plot demonstrated an unsymmetrical shape (Fig.3). In addition, the subsequent

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Egger's test shown $P<.05$ ($P=0.025$), which indicated a possible publication bias existing. Sensitivity analysis was conducted by the leave-one-out method. For the spending, we found that the I^2 value decreased to 0% after excluding the trial conducted by Li et al (2018) [19], however, the pooled result was stable by using sensitively analysis. Li et al reported that they performed retroperitoneal catheter drainage, thus, we believed that the high heterogeneity may raised from following factors: sample size, the quality of trial and different PCD pathway.

4. Discussion

The present systematic review and meta-analysis identified three RCTs and five cohort studies investigated the effect and safety of APD in treating SAP patients. This analysis found that compared with the conventional "step-up" treatment, APD significantly decreased all-cause mortality, the incidence of organ failure, the length of hospital stays and the spending during hospitalization, but did not increase the risk of extra infectious. However, there were some differences between cohort studies and RCTs subgroup in the aspect of the incidence of organ failure.

Peritoneal fluid formation was contributed by two reasons. Firstly, the host cytokine cascades activated by local inflammation of the pancreas appears in the first week or even the second[2]. Excessive inflammatory factors aggravated pancreatic vasospasm and capillary permeability, which resulted in the massive plasma extravasation and extensive cell necrosis. Secondly, a large mount of intravenous infusion was used to correct the insufficient of circulating volume and electrolyte disturbance in acute pancreatitis patients, which leaded to the accumulation of fluid in the peritoneal cavity that was capillary leak syndrome (CLS). And the leakage of pancreatic fluid would also accelerate the increase of ascites. The pancreatic ascites in the early stage of acute pancreatic adenitis was considered to be bacteria-free[22], however, ascites was rich in proteases, lipases, and cytokines, it would be a natural medium for the certain intestinal bacteria.

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Hence, it is important to clean the inflammatory factors in the pancreatic ascites. Lourenilson et al found that peritoneal lavage could let to a decrease in serum levels of TNF-α and IL-6 and an increase in IL-10[23] in the rat model of SAP. And, another retrospectively clinical trial also concluded that peritoneal lavage reduced the mortality and incidence of related complication [24]. However, two meta-analysis showed opposite conclusions, which had high risk of bias and lack further exploration of infectious incidence[25,26]. Comparing with peritoneal lavage, a recent RCT demonstrated that there were no difference in mortality, complication and the risk of infection in PCD group[27]. The main difference between two treatments was that PCD could better reduce intra-abdominal hypertension. PCD usually performed after the formation of pancreatic pseudocyst which needed 4 weeks. Even PCD was an early intervention and reduced the surgery-caused risk.. APD was recommended as a better option which needed only 2 weeks. Plenty of studies demonstrated a significant decrease in serum IL-6, TNF-α and C-reaction protein levels in APD group compared with traditional "step-up" therapy group patients[10,20]. Moreover, Liang et al found that APD reduced the incidence of further minimally invasive necrectomy[16]. We believe that the implementation of APD in the early phase significantly reduced the levels of cytokines and restrained the possibility of endogenous bacterial growth, which would helpful for avoiding disease from getting worsen and contributed to a decline in all-cause mortality during hospitalization eventually.

It is worth noting that another major reason for the decline in mortality may be a relieving of intra-abdominal pressure (IAP) through APD. The increase in IAP could be due to inflammatory response and CLS-induced abdominal effusion and visceral edema[28]. Increased IAP reduced the intestinal mucosal perfusion leading to the ischemia of tissues and destroyed the function of intestinal mucosal barrier, consequently increased the possibility of bacterial translocation. And increased IAP also had some adverse effects on the host circulatory system reflected in the reduction of venous return and cardiac output. There

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fore, early implement of APD was proved to be necessary.

As an invasive intervention, operator usually focuses on whether APD brings extra risks of infectious c omplications, because the impact of exogenous infection has a fatal impact on critical patients and can offset the endeavor by previous treatment. In this present study, we found that APD did not increase th e risk of microbial infection, sepsis and bacteremia when compared with PCD.

This trial was the first systematic review comparing APD with conventional "step-up" method. And we acknowledged that there were some limitations included in this review. Firstly, this study relied greatly on cohort studies due to limited number of published RCTs, but it was generally known that observatio nal studies exist the risk of selection bias, although theses cohort studies were high quality. Secondly, despite sensitivity analysis proved the stability of results of spending, the heterogeneity of it was very l arge, so we should keep cautious about this result. Thirdly, the effect of APD could be influenced by t he difference in catheter position, catheter lumen, catheter time and even diameter, but there were insuf ficient subgroups to do this. Lastly, all included studies had small sample size. In meta-analysis, the sm all study effect in small trials could magnify the positive effects of the intervention compared with larg e trials [29]. On another hand, the insufficient in component studies and their small sample size could p artly explained the difference between the results of the cohort studies and RCTs subgroups. Hence, ca ution must be practiced when comprehend the final pooled results, even we have improved the search s trategy, and performed according to PRISMA statement strictly.

Nowadays, there are no relevant guidelines or consensus to recommend APD for curing of AP, howeve r, its potential clinical practice value deserves our continuous attention. In the further trials, researcher a nd clinicians should focus on the following points. First, which type of patients may benefit from APD? Second, it may be interesting to investigate the best time for the implement of APD from onset. Finall

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y, it is necessary to monitor the change of intra-abdominal pressures continuously during the curing of AP, because that is important for exploring the reasons of effectiveness about APD.

5. Conclusions

In conclusion, the current evidence found that APD could reduce the all-cause mortality, the rate of organ failure, length of hospital stay and spending during hospitalization while did not increase the extra risk of infectious complications. APD may be a promising option for the early trea tment of AP, however, the results largely relied on cohort studies, which contain deficiency in methodological quality. Hence, further study of APD for the AP curing is warranted in high-quality and large-scale RCTs.

Author contributions

Zongqing Lu and Xingxing Zhu contributed equally to this work, who designed the study, collected and analysed data, drafted and revised of the article, and final approved the version to be published; Min Yang and Tian-feng Hua contributed to interpretation of data, revising of the article and final approval of the version to be published; Wenyan Xiao, Jin Zhang and Di Jia contributed to acquisition of data, analysis data, drafting the article.

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Figure Legends:

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Figure 1: Flow diagram of study selection.

Figure 2: Forest plot for all-cause mortality.

Figure 3: Funnel plot for all-cause mortality.

Supplementary Figure 1: Forest plot for the rate of organ failure.

Supplementary Figure 2: Forest plot for the risk of infectious complications.

Supplementary Figure 3: Forest plot for the length of hospital stay.

Supplementary Figure 4: Forest plot for spending

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Table 1 The characteristics of included trials comparing Abdominal Paracentesis Drainage with conventional "set-up" approach.

First author (publish year)	Study period	Study type	Sample size		Mean age (years)		Etiology of AP	APACHE Ⅱ at admission		Catheter diameter	Outcomes[#]
			APD group	Non-APD group	APD group	Non-APD group		APD group	Non-APD group		
Wei-hui Liu[10] (2015)	2009.06-2011.06	RCS	53	49	59±13.5	57±12.7	Gallstone, alcohol abuse, hyperlipemia	15.5±3.55	14.7±3.25	8F-22F	1,2,3,4,5,6,7
Yin Li[19] (2018)	2012.01-2017.12	RCS	46	42	47.0±14.6	48.8±13.5	NR	NR	NR	8F-16F	1,2,5,6,7
Li-ye Liu[15] (2015)	2013.08-2014.10	PCS	126	129	57±13	55±11	Biliary, alcohol abuse	13.5±3.1	14.2±2.8	8F-22F	1,3,4
Jun-wei Ma[20] (2017)	2009.11-2015.11	RCS	43	39	55.89±4.59	56.28±3.60	Biliary, alcohol abuse	23.69±2.46	22.95±3.26	NR	1,3,4,5,6,7
Zhu Huang[17] (2016)	2010.05-2015.05	RCS	68	64	50 .33±11.17	48.79±9.54	non- HTG induced	15.21 ± 3.48	15.47 ± 3.10	NR	1,6
Wen-ting Luo[18] (2019)	2016.07-2018.11	RCT	49	49	45.69±11.87	44.37± 11.21	NR	17.51±4.47	17.86±4.64	12F	2,6
Hong-yin Liang[16] (2016)	2015.01-2016.04	RCT	83	78	55.6±14.6	53.7±12.8	Gallstone, alcohol abuse, hyperlipemia	13.4±4.1	13.9±4.9	8F-16F	1,2,5,6
Liang Zhang[21] (2020)	2014.03-2018.11	RCT	84	84	63.22±12.43	63.43±12.37	Biliary, alcohol abuse	NR	NR	NR	2,6,7

APD=Abdominal paracentesis drainage; AP=Acute pancreatitis; NR=Not report; RCS=Retrospective cohort study; PCS=Prospective cohort study; RCT=Randomized control trial; HTG=hypertriglyceridemia

[#]Outcomes: 1=all-cause mortality; 2=the rate of organ failure; 3=the rate of sepsis; 4=the rate of bacteremia; 5=spending; 6=length of hospital stay; 7=the rate of microbial

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infection

Data are mean ± standard deviation unless otherwise indicated

Table 2 The risk of bias assessment of included cohort studies.

First author (publish year)	Clauses								Score	Grade
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)		
Wei-hui Liu (2015)	1	1	1	0	1	1	1	1	7	High quality
Yin Li (2018)	1	1	1	0	1	1	1	1	7	High quality
Li-ye Liu (2015)	1	1	1	1	1	1	1	1	8	High quality
Jun-wei Ma (2017)	1	1	1	0	1	1	1	1	7	High quality
Zhu Huang (2016)	1	1	1	0	1	1	1	1	7	High quality

(1)=representativeness of the exposed cohort; (2)=selection method for non-exposed cohort; (3)=methods for exposure factors determined; (4)=demonstration that the outcome was not present at the beginning of the trial; (5)=the most important confounding factors or (and) any other confounding factors has been controlled; (6)=adequate assessment of the outcome related; (7)=adequate duration of follow-up; (8)=adequate follow-up in each cohort.

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Table 3 The risk of bias assessment of included randomized control trials.

First author (publish year)	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other bias	Overall Risk of bias
Wen-ting Luo (2019)	Coin toss generation	Unclear	Single-blinding	Yes	No	No	No	Unclear risk of bias
Hong-yin Liang (2016)	Computer generation	opaque envelopes	Single-blinding	Yes	No	No	No	Low risk of bias
Liang Zhang (2020)	randomized digital table generation	Unclear	Single-blinding	Yes	No	No	No	Unclear risk of bias

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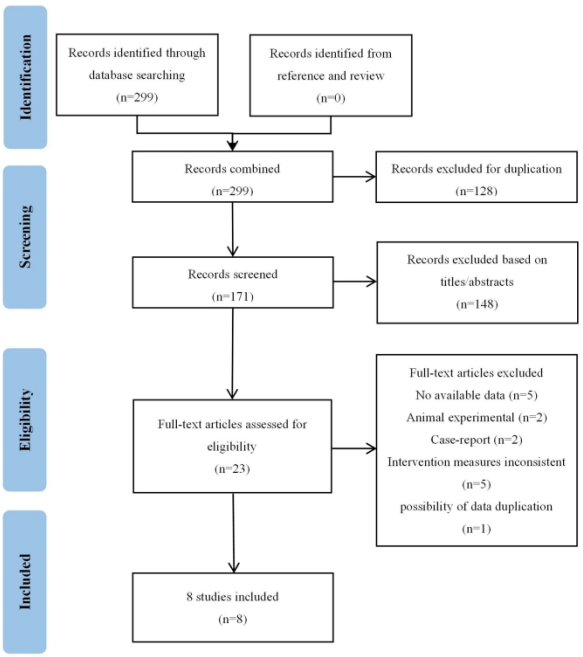


Figure 1 Flow diagram of study selection.

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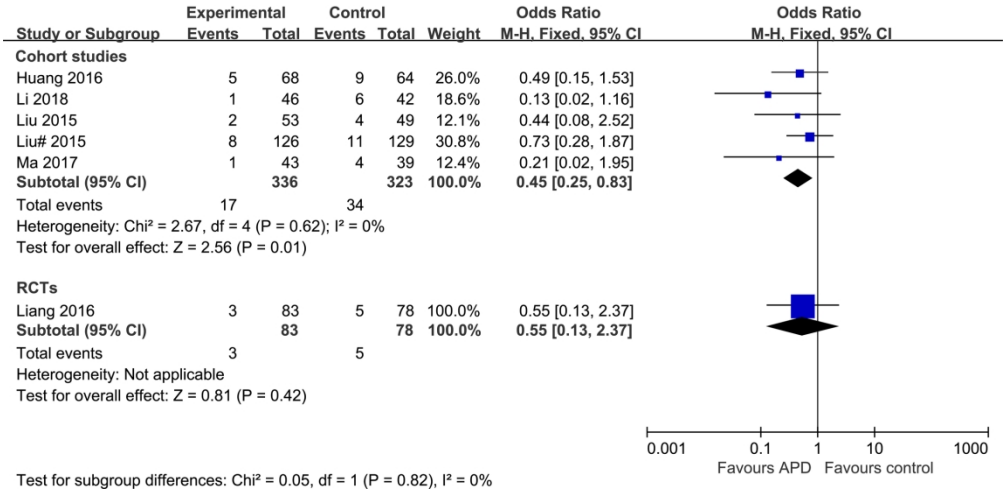


Figure 2 Forest plot for all-cause mortality

195x96mm (300 x 300 DPI)

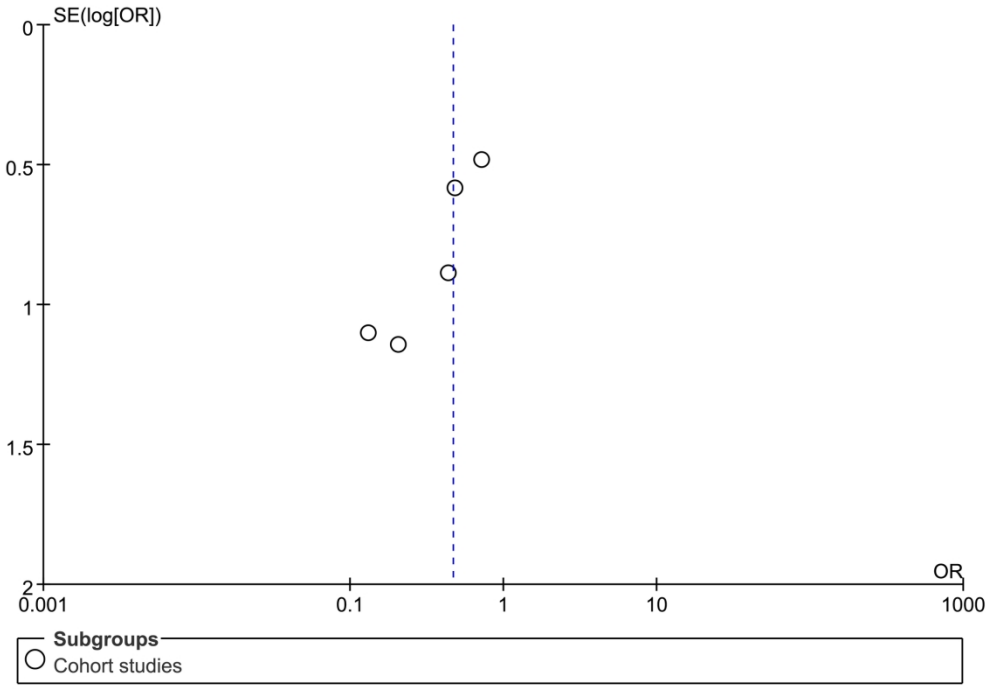


Figure 3 Funnel plot for all-cause mortality.

195x139mm (300 x 300 DPI)

Supplementary Table 1: PubMed search Strategy.

PubMed
#1 “acute pancreatitis” OR “severe acute pancreatitis”
#2 AP OR SAP
#3 Pancreatitis (Mesh)
#4 #1 OR #2 OR #3
#5“Abdominal Paracentesis Drainage” OR APD
#6 “step-up approach”
#7 #5 OR #6
#8 randomized controlled trial
#9 controlled clinical trial
#10 cohort study
#11 RCT OR “observational study” OR “cohort study” OR cohort
#12 #8 OR #9 OR #10 OR #11
#13 #4 AND #7 AND #12

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Supplementary Table 2 GRADE evidence of each predefined outcome.

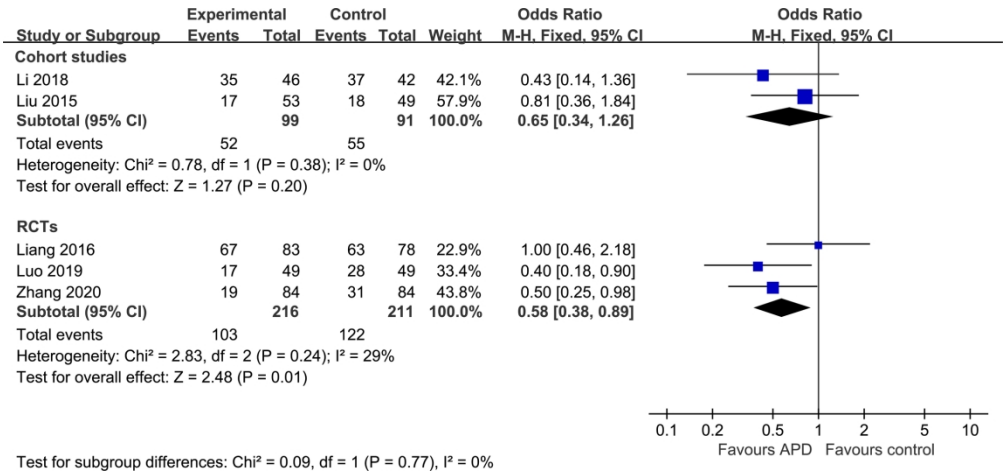
Outcome	Study type	No. study	Sample size (E/C)	Inconsistency	Indirectness	Imprecision	Large effect	Confounder	Dose-response gradient	GRADE Evidence	Outcome Importance
All-cause Mortality	RCT	1	83/78		Directness	High Imprecise				LOW	Critical
	Cohort study	5	336/323	Consistency	Directness	Imprecise	Large	NO	NO	LOW	
Microbial Infection	RCT	1	84/84		Directness	High Imprecise				Very LOW	Critical
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	LOW	
Sepsis	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Bacteremia	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Organ Failure	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	2	99/91	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	
Length of hospital stay	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	4	210/194	Consistency	Directness	Imprecise	NO	NO	NO	LOW	

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Spending	RCT	2	167/162	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	

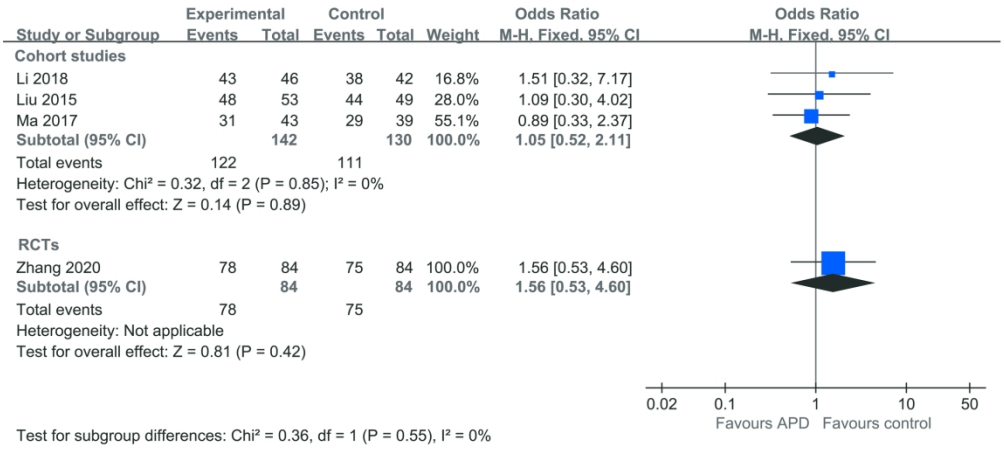
E=experimental group; C=Control group; RCT=Randomized control trial

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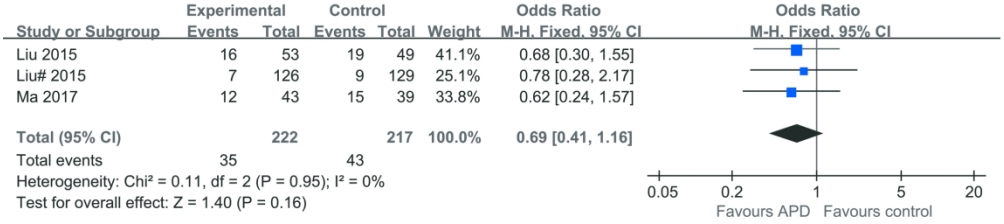


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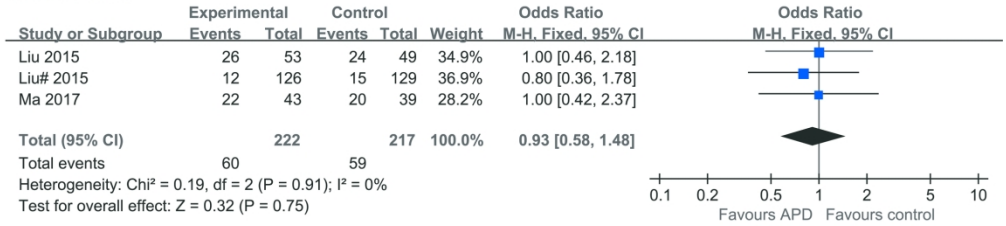
a: microbial infection



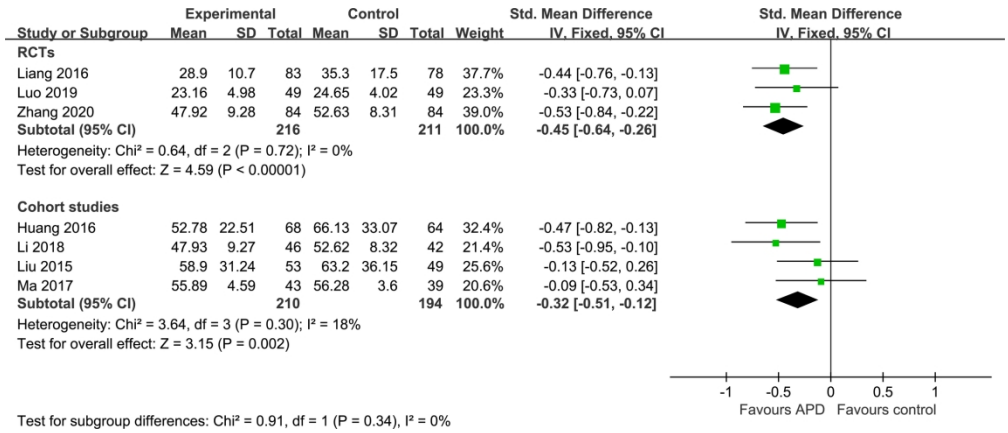
b: sepsis



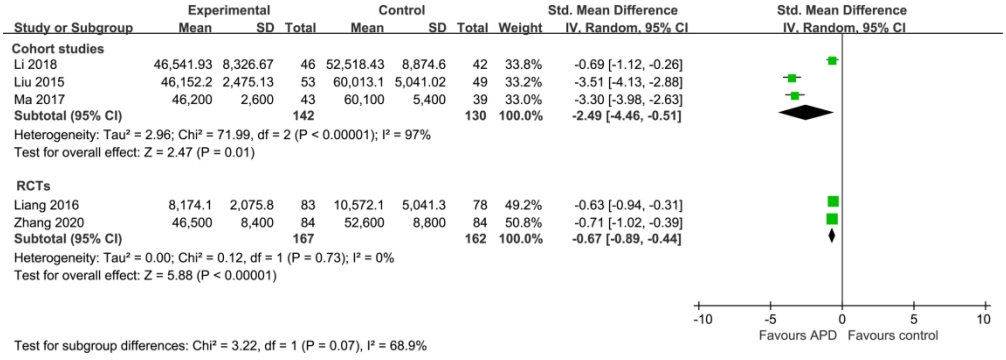
c: bacteremia



195x224mm (300 x 300 DPI)



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195x69mm (300 x 300 DPI)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11-14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis

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Primary Subject Heading:	Intensive care
Secondary Subject Heading:	Gastroenterology and hepatology, Emergency medicine
Keywords:	Pancreatic disease < GASTROENTEROLOGY, Adult intensive & critical care < INTENSIVE & CRITICAL CARE, Minimally invasive surgery < GYNAECOLOGY

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Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis

Zongqing Lu[1,2] , Xingxing Zhu[1,2], Tian-feng Hua[1,2], Jin Zhang[1,2], Wenyan Xiao[1,2], Di Jia[1,2], Min Yang[1,2*]

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Conflicts of Interest

The authors declare that there is no conflict of interest

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Abbreviations

AP: Acute pancreatitis

APD: Abdominal paracentesis drainage

PCD: Percutaneous catheter drainage

MAP: Mild acute pancreatitis

MSAP: Moderate severe acute pancreatitis

SAP: Severe acute pancreatitis

RIS: Required information size

CLS: Capillary leak syndrome

IAP: Intra-abdominal pressure

RCT: Randomized controlled trial

OR: Odds ratio

CI: Confidence interval

SMD: Standardized mean difference

CNKI: China National Knowledge Infrastructure

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Abstract

Objectives

This systematic review and meta-analysis were performed to investigate whether abdominal paracentesis drainage (APD) is more advantageous in treating acute pancreatitis (AP) patients when compared with conventional percutaneous catheter drainage (PCD), and evaluated the risk of infectious complications caused by APD.

Design

Systematic review and meta-analysis.

Methods

PubMed, EMBASE, Cochrane Library, OVID, CNKI and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) from inception to July 25, 2020. Studies related to comparing APD and conventional PCD method for curing AP patients were included.

Outcomes

The primary outcome was all-cause mortality. The secondary outcomes were the rate of organ failure, infectious complications, hospitalization expenses and length of hospital stay.

Results

A total of five cohort studies and three RCTs were included. Compared with conventional PCD method, pooled results suggested that APD significantly decreased all-cause mortality (Cohort studies: OR 0.48, 95%CI 0.26-0.89, P=0.02), length of hospital stay (Cohort studies: SMD -0.31, 95%CI -0.53, -0.10, P=0.005; RCTs: SMD -0.45, 95%CI -0.64, -0.26, P<0.001) and hospitalization expenses (Cohort studies: SMD -2.49, 95%CI -4.46, -0.51, P<0.001; RCTs: SMD -0.67, 95%CI -0.89, -0.44, P<0.001). In addition, no evidence proved that APD increased the incidence of infectious complications. However, there was some difference between cohort

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studies and RCTs subgroup in the incidence of organ failure (Cohort studies: OR 0.66, 95%CI 0.34-1.28, P=0.22; RCTs: OR 0.58, 95%CI 0.35-0.98, P=0.04).

Conclusions

Currently, limited evidence shows that APD might decrease all-cause mortality, expenses and the length of stay on AP patients during hospitalization, while the extra infectious risk was not increased. This study provided new insight into treating AP. However, these final meta-analysis results must be interpreted very cautiously since a larger dependence on observational trials in this study.

KEYWORDS: abdominal paracentesis drainage; acute pancreatitis; percutaneous catheter drainage; systematic review; meta-analysis

Strengths and limitations of this study

1. This trial was the first systematic review and meta-analysis comparing abdominal paracentesis drainage (APD) with conventional percutaneous catheter drainage (PCD) method in treating acute pancreatitis (AP) patients.

2. Thorough searching of six major electronic databases, including two major non-English language databases.

3. The study protocol has been registered in PROSPERO and reported according to PRISMA guidelines.

4. This study relied greatly on cohort studies due to the limited number of published RCTs, which means that final pooled results must be interpreted very cautiously.

5. The meta-analysis results might be affected by the length of follow-up, canal diameter of APD and etiology given the limited numbers of studies. Confirmation of findings by further clinical trials is warranted.

1. Introduction

Acute pancreatitis (AP), characterized by rapid progression, severe complications and high mortality, is one of the most common diseases causing admission to the intensive care units [1]. AP was divided into acute

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pancreatitis (MAP), moderate severe acute pancreatitis (MSAP) and severe acute pancreatitis (SAP) according to the updated 2012 Atlanta Classification, MAP only have upper abdominal pain, nevertheless local or systemic complications, such as peripancreatic collection, will occur in MSAP, SAP stage, and persistent organ failure eventually [2].

Currently, the treatment of AP has undergone a winding exploration for a long history. In the past, conservative treatments including active rehydration and high-dose antibiotics have been used in early stage, necrosectomy was performed when previous treatment would not effectively prevent deterioration [3]. However, open surgery was associated with high mortality, infectious complications and prolonged recovery time[4]. In 1998, Freeny et al firstly reported a novel treatment, which is commonly known as percutaneous catheter drainage (PCD), then the efficacy and safety of which were confirmed by many clinical trials [5]. Since then, the AP treatment period has entered a PCD-based "step-up" time [6,7], the purpose of which was aimed at controlling sources of infection progressively and reasonably instead of removing necrosis immediately, thereby reducing the risk of increased death caused by traumatic surgery[8]. However, this conventional PCD-based "step-up" treatment could not meet the earlier clinical requirements, because PCD must be performed at least 4 weeks after onset when the collection of walled-off necrosis was formed [9]. However, in the early stage of AP, the release of a large number of inflammatory mediators leads to an increase in tissue capillary permeability, hence, the fluid is generated and dissociated into the abdominal cavity or even the pelvic cavity. These toxic substances destructed the intestinal mucosal barrier and then be absorbed into the blood through the peritoneum, causing inflammatory cascades, in hence, aggravating systemic inflammatory response syndrome (SIRS). Recently, a retrospective cohort study reported a new insert catheters technique, called abdominal paracentesis drainage (APD), which was carried out through right paracolic sulci or left paracolic sulci, when the abdominal collection volume was greater than 50ml [10]. The results of this study showed that early APD treatment could effectively reduce the release of

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inflammatory factors and improve the clinical prognosis. However, the controversy has mainly focused on whether the treatment process would bring exogenous infection or not.

Therefore, we systematically reviewed available articles to investigate whether the APD could significantly decrease mortality when compared with the conventional "step-up" method in treating AP patients. The secondary objectives were to explore the risk of infectious complications, hospitalization expenses and length of hospital stay.

2. Material and Methods

The present systematic review and meta-analysis were reported in accordance to the Preferred Reporting item for Systematic Review and Meta-analysis (PRISMA) statement [11]. And, this review has been registered on the PROSPERO website as No. CRD42020168537.

2.1 Data sources and searches

PubMed, EMBASE, Cochrane Library, OVID, China National Knowledge Infrastructure (CNKI) and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) about the abdominal paracentesis drainage for curing AP patients from inception to February 15, 2020, by using keywords combined with Medical Subject Headings (MeSH) terms, regardless of language and region. We retrieved it again on July 25, 2020, to included newly published paper. In addition, the reference lists of related literature were manually searched for possible trials. The search strategy for PubMed was shown in Supplementary Table 1.

2.2 Study selection

Zongqing Lu and Xingxing Zhu searched independently, according to predefined inclusion and exclusion criteria. Duplicate literature deletion, title and abstract screening for relevance had been done using Endnote software. Then, the full text was acquired to determine inclusion eligibility eventually. Any disagreement was resolved

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through discussion, a third review author (Min Yang) participated where necessary.

Published RCTs and cohort studies meeting the following criteria were included: 1.population: Adult patients with acute pancreatitis regardless of pathogenesis, however, detailed diagnostic criteria for acute pancreatitis must be required, and each participant must conform to the requirement of APD: 1) enough volume of celiac fluid collections (> 50mL); 2) a feasible pathway existed. No limitation on race and nationality. 2.intervention: APD was administered within 4 weeks of onset. 3.comparison: traditional "step-up strategy" in which APD was not performed. 4.the sample size of each group exceeds 20.

2.3 Outcomes and Data Extraction

Two authors (Zongqing Lu and Di Jia) independently extracted data using a pre-piloted form designed by Excel 2019 software (Microsoft Corporation), and the results were confirmed by another author (Tianfeng Hua). The collected data include: the first author, publish year, sample size, mean age and sex ratio of each group, study period, following time, cause of AP, APACHE Ⅱ scores at admission, catheter diameter and outcomes data. If any information above was inadequate, we contacted the original author via email to consult related data. We resolved discrepancies through discussion. The predefined primary outcomes were all-cause mortality during hospitalization. The secondary outcomes were the rate of organ failure, the rate of related infectious complications such as microbial infection, sepsis and bacteremia after treatment, spending during hospitalization and the length of hospital stay.

2.4 Quality Assessment

The quality of filtered articles has been assessed by two authors (Wenyan Xiao and Jin Zhang) respectively. The risk of bias tools performed were specific to the study type. For RCTs, the revised Cochrane tool to assess risk of bias (ROB-2) was used for each of the following five domains: randomization process, deviations from intended interventions, missing outcomes data, measurement of the outcome and selection of the reported results [12]. We

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reviewed each RCT and classifies them as low, probably low, probably high or high risk of bias. However, blinding of patients and clinicians was often unrealistic in invasive treatment, and which has limited influence on primary outcomes. So, trials were seen as low risk in the domain of deviations from intended interventions and measurement of the outcome even without double-blinding.

Meanwhile, cohort studies were assessed using the Risk Of Bias In Non-randomised Studies - of Interventions tool (ROBINS-I), in which seven key domains were contained: 1.confounding; 2.selection of participants; 3.classification of interventions; 4.deviations from intended interventions; 5.missing data; 6.measurement of outcomes; 7.selection of the reported results It was widely used in the non-RCT evaluation, and recommended by the Cochrane collaboration [13].

2.5 Quality of evidence

Two authors assess the quality of each evidence respectively by using the GRADE system (Grading of Recommendations Assessment, Development and Evaluation) for risk of bias, inconsistency, indirectness, imprecision, publication bias, large effect, confounder and dose-response gradient [14]. The quality was divided into very low, low, moderate, or high. The results were generated by using the GRADE Profiler.

2.6 Statistical Analysis

RevMan 5.3 (Nordic Cochrane Centre) was used for the analysis of the included studies. Odds ratio (OR) and the 95% confidence intervals (CIs) were calculated for dichotomous data by using the Mantel-Haenszel method. For continuous data, standard mean difference (SMD) and 95%CIs were calculated, due to the differences in measurement methods and units. *P* values less than 0.05 were considered to be significant. Meta-analysis was performed if predefined outcomes from two or more studies. Statistical heterogeneity across trials was examined using the I^2 statistical tests as well as *P* values. Those with $P<0.1$ and I^2 greater than 50% were seen as significant heterogeneity. However, due to the limited number of component studies and patients, the statistical

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test for heterogeneity maybe not sensitive. At meanwhile, the fix-effects model could amplify the weight of large sample studies. Thus, we used a random-effects model to merge each result. Then subgroup analysis according to the RCTs or cohort studies were performed when sufficient data available, to avoid inappropriate pooled results influence the validity of evidence. The sensitivity analysis was carried out by leave-one-out method, to explore the sources of heterogeneity and examined the stability of results. We performed a funnel plot and Egger's test to detect the possible publication bias.

3. Results

3.1 Literature Research

The flow diagram showed the process of literature screening, selection and reasons for exclusion (Fig.1). Our initial search yielded 299 records. After removing duplications and reviewing the titles/abstracts, 23 articles were considered to be potentially eligible for inclusion. After reading the full-text, 15 studies were excluded for the following reasons: no available data (n=5), animal experimental (n=2), case-report (n=2), intervention measures inconsistent (n=5) and the possibility of data duplication (n=1). As a result, eight studies [10,15-21] were eventually included in this meta-analysis.

3.2 Trials Characteristics

The characteristics of the eligible trials had been summarized in Table 1. There were five cohort studies and three RCTs published between 2014 and 2020, which compared the clinical prognosis of AP patients who were cured by APD or conventional "step-up" approach. Population sizes ranged from 82 to 255, with a total of 1086 patients. Except for the catheter diameter and the catheter numbers, there was no statistical difference of baseline between APD and control groups in each trial, according to the report.

3.3 Risk of Bias Assessment and GRADE Profile Evidence

The details about the risk of bias were respectively shown in Table 2 and Figure 2. For included cohort studies,

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three studies were classified as low risk of bias [10, 15, 17], *Li et al* were categorized as a severe risk of bias due to the lack of necessary baseline comparison and we could not determine whether the outcome indicators were measured at the same time of each group in this trial[19]. Meanwhile, *Ma et al* did not report an appropriate method to balance the treatment start time, so we categorize it as moderate risk through discussion [20]. For RCTs included, one was categorized as low risk of bias in each domain [16], others were considered as some concerns because they did not report the way of allocation concealment [18,21].

GRADE evidence profiles are showed in Supplementary Table 2. Due to either unclear risk of bias or imprecision resulted from small sample size influence and relative wide confidence interval, the quality of evidence of outcomes in RCTs subgroup was low or very low, according to the GRADE framework. In addition, we also determined the GRADE evidence of cohort studies subgroup as low or very low eventually, considering the natural risk of observational trials, and small effect size except for all-cause mortality.

3.4 Primary outcome

All-cause mortality during hospitalization was reported in six trials, which included 820 patients [10,15-17,19,20]. Compared with the conventional "step-up" approach, five cohort studies [10,15,17,19,20] of APD intervention reported a significantly reduced risk of all-cause mortality in acute pancreatitis patients (OR 0.48, 95%CI 0.26-0.89, P=0.02) with low heterogeneity (I^2=0.0%) (Fig.3). However, there was no significantly differences in all-cause mortality reported by another RCT (n=161, OR 0.55, 99%CI 0.13-2.37, P=0.42) [16].

3.5 Secondary outcomes:

3.5.1 The rate of organ failure: Five studies provided data on the rate of organ failure with totaling 617 patients [10,16,18,19,21]. Compared with conventional "step-up" approach, the results of cohort studies subgroup [10,19] shown that there were no significant differences in the incidence of organ failure (n=190, OR 0.66, 95%CI 0.34-1.28, P=0.22). However, the results of RCTs subgroup [16,18,21] supported APD (n=427, OR 0.58, 95%CI 0.38-0.98,

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P=0.04) (Supplementary Figure 1). We considered that the inconsistencies of results between two subgroups were resulted from the difference in sample size because preferred APD trends have appeared in cohort studies subgroup. The heterogeneity in cohort studies and RCTs subgroup was I^2=0% and I^2=29%, respectively.

3.5.2 The risk of infectious complications: Five studies reported the number of infectious complications, which include microbial infection [10,19-21], sepsis [10,15,20] and bacteremia [10,15,20], with totaling 695 patients. We pooled data from three cohort studies[10,19,20]which showed no significant differences in the incidence of microbial infection (OR 1.05, 95%CI 0.52-2.12, *P*=0.89) with I^2=0%, and this was consistent with another RCT's result (OR 1.56, 95%CI 0.53-4.60, *P*=0.42)[21]. For the incidence of sepsis and bacteremia, the pooled results from three cohort studies showed that APD could not significantly increase the related risk versus the conventional "step-up" approach (sepsis: OR 0.69, 95%CI 0.41-1.16, *P*=0.16; bacteremia: OR 0.93, 95%CI 0.58-1.48, *P*=0.75). The heterogeneity in both of them was I^2=0%. (Supplementary Figure 2)

3.5.3 Length of hospital stay: Four cohort studies (totaling 404 patients) [10,17,19,20] and three RCTs (totaling 427 patients) [16,18,21] were included in this analysis. The results of cohort studies subgroup showed that APD significantly decreased the length of hospital stay when compared with conventional "step-up" approach (SMD -0.31, 95%CI -0.53, -0.10, *P*=0.005). In addition, the pooled results from three RCTs also supported APD (SMD -0.45, 95%CI -0.64, -0.26, *P*<0.001). The heterogeneity in each subgroup was I^2=18% and I^2=0%. (Supplementary Figure 3).

3.5.4 Spending: We extracted the data about spending during hospitalization from three cohort studies and two RCTs, totaling 601 patients [10,16,19-21]. The pooled results of cohort studies subgroup [10,19,20] demonstrated that APD could significantly reduce the spending versus conventional "step-up" approach (SMD -2.49, 95%CI -4.46, -0.51, *P*<0.001), which was consistent with the results of RCTs subgroup [16,21] (SMD -0.67, 95%CI -0.89, -0.44, *P*<0.001), and heterogeneity were I^2=97%, I^2=0% in cohort studies and RCTs subgroup respectively.

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(Supplementary Figure 4)

3.6 Publication Bias and Sensitivity Analysis

We just included cohort studies to explore the potential publication bias for primary outcomes, due to the limited number of RCT. The funnel plot demonstrated an unsymmetrical shape (Fig.4). In addition, the subsequent Egger's test shown $P<.05$ ($P=0.025$), which indicated a possible publication bias exists. Sensitivity analysis was conducted by the leave-one-out method. For the spending, we found that the I^2 value decreased to 0% after excluding the trial conducted by Li et al (2018) [19], however, the pooled result was stable by using sensitivity analysis. Li et al reported that they performed retroperitoneal catheter drainage, thus, we believed that the high heterogeneity may raise from following factors: sample size, the quality of trial and different PCD pathways. At meanwhile, we excluded studies with a severe risk of bias and finally proved that the pooled results were still stable.

4. Discussion

The present systematic review and meta-analysis identified three RCTs and five cohort studies that investigated the effect and safety of APD in treating SAP patients. This pooled results of cohort studies found that compared with the conventional "step-up" treatment, APD could significantly decrease all-cause mortality, the incidence of organ failure, the length of hospital stays and the spending during hospitalization, but did not increase the risk of extra infectious. However, the analysis of RCTs subgroup presented differently in the aspect of the incidence of organ failure.

The peritoneal fluid formation was contributed by two reasons. Firstly, the host cytokine cascades activated by local inflammation of the pancreas appear in the first week or even the second [2]. Excessive inflammatory factors aggravated pancreatic vasospasm and capillary permeability, which resulted in massive plasma extravasation and extensive cell necrosis. Secondly, a large amount of intravenous infusion was used to correct

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the insufficient circulating volume and electrolyte disturbance in acute pancreatitis patients, which led to the accumulation of fluid in the peritoneal cavity that was capillary leak syndrome (CLS). And the leakage of pancreatic fluid would also accelerate the increase of ascites. The pancreatic ascites in the early stage of acute pancreatic adenitis were considered to be bacteria-free [22], however, ascites were rich in proteases, lipases, and cytokines, which would be a natural medium for certaining intestinal bacteria.

Hence, it is important to clean the inflammatory factors in the pancreatic ascites. Lourenilson et al found that peritoneal lavage could lead to a decrease in serum levels of TNF-α and IL-6 and an increase in IL-10 [23] in the rat model of SAP. And, another retrospectively clinical trial also concluded that peritoneal lavage reduced the mortality and incidence of related complications [24]. However, two meta-analysis showed opposite conclusions, which had a high risk of bias and lack further exploration of infectious incidence [25,26]. Comparing with peritoneal lavage, a recent RCT demonstrated that there were no differences in mortality, complication and the risk of infection in PCD group [27]. The main difference between two treatments was that PCD could better reduce intra-abdominal hypertension. PCD is usually performed after the formation of pancreatic pseudocyst which needed 4 weeks. Even PCD was an early intervention and reduced the surgery-caused risk. APD was recommended as a better option which needed only 2 weeks. Plenty of studies demonstrated a significant decrease in serum IL-6, TNF-α and C-reaction protein levels in APD group compared with traditional “step-up” therapy group patients [10,20]. Moreover, Liang et al found that APD reduced the incidence of further minimally invasive necrosectomy [16]. We believe that the implementation of APD in the early phase significantly reduced the levels of cytokines and restrained the possibility of endogenous bacterial growth, which would helpful for avoiding disease from getting worsen and contributed to a decline in all-cause mortality during hospitalization eventually.

It is worth noting that another major reason for the decline in mortality may be a relieving of intra-abdominal

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pressure (IAP) through APD. The increase in IAP could be due to inflammatory response and CLS-induced abdominal effusion and visceral edema [28]. Increased IAP reduced the intestinal mucosal perfusion leading to the ischemia of tissues and destroyed the function of intestinal mucosal barrier, consequently increased the possibility of bacterial translocation. And increased IAP also had some adverse effects on the host circulatory system reflected in the reduction of venous return and cardiac output. Therefore, early implementation of APD was proved to be necessary.

As an invasive intervention, the operator usually focuses on whether APD brings extra risks of infectious complications, because the impact of exogenous infection has a fatal impact on critical patients and can offset the endeavor by previous treatment. According to the analysis of cohort studies, we found that APD did not increase the risk of microbial infection, sepsis and bacteremia when compared with PCD. These results were still stable even trials with a severe high risk of bias excluded.

This trial was the first systematic review comparing APD with conventional "step-up" method. And we acknowledged that there were some limitations included in this review. Firstly, this study relied greatly on cohort studies due to a limited number of published RCTs, but it was generally known that observational studies exist the risk of selection bias. Secondly, despite sensitivity analysis proved the stability of results of spending, the heterogeneity of it was very large, so we should keep cautious about this result. Thirdly, the effect of APD could be influenced by the difference in catheter position, catheter lumen, catheter time even diameter. And the rate of organ failure and related infectious complications might be affected by the length of following time, but there were insufficient subgroups to do this. Lastly, all included studies had small sample size. In meta-analysis, the small study effect in small trials could magnify the positive effects of the intervention compared with large trials [29]. On another hand, the insufficient in component studies and their small sample size could partly explain the difference between the results of the cohort studies and RCTs subgroups. Hence, caution must be practiced when

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comprehending the final pooled results, even we have improved the search strategy, and performed according to the PRISMA statement strictly.

Nowadays, there are no relevant guidelines or consensus to recommend APD for curing of AP, however, its potential clinical practice value deserves our continuous attention. In further trials, researchers and clinicians should focus on the following points. First, which type of patients may benefit from APD? Second, it may be interesting to investigate the best time for the implementation of APD from onset. Finally, it is necessary to monitor the change of intra-abdominal pressures continuously during the curing of AP, because that is important for exploring the reasons of the effectiveness of APD.

5. Conclusions

Currently, limited cohort studies presented that APD might decreased all-cause mortality, expenses and the length of stay on AP patients during hospitalization while extra infectious risk was not increased. This study provided new insight into treating AP. However, we should remain circumspection and expect more related high-quality RCTs since these final pooled results are essentially based on observational studies.

Author contributions

Zongqing Lu and Xingxing Zhu contributed equally to this work, who designed the study, collected and analyzed data, drafted and revised the article, and final approved the version to be published; Min Yang and Tian-feng Hua contributed to the interpretation of data, revising of the article and final approval of the version to be published; Wenyan Xiao, Jin Zhang and Di Jia contributed to acquisition of data, analysis data, drafting the article.

Data availability statement

Data are available upon reasonable request. All data relevant to the study are included in the article or uploaded as supplementary information. The data set used and analysed for this review will be available from the

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corresponding author upon a reasonable request.

Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Ethics Statement

This study did not require ethical approval as the data used have been published previously, and hence are already in the public domain.

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Figure Legends:

Figure 1: Flow diagram of study selection.

Figure 2: The risk of bias about included RCTs by using ROB-2 tool, A: traffic light plot of RCT bias assessment; B: weighted summary plot of the overall type of bias encountered in RCTs

Figure 3: Forest plot for all-cause mortality.

Figure 4: Funnel plot for all-cause mortality.

Supplementary Figure 1: Forest plot for the rate of organ failure.

Supplementary Figure 2: Forest plot for the risk of infectious complications.

Supplementary Figure 3: Forest plot for the length of hospital stay.

Supplementary Figure 4: Forest plot for spending.

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Table 1 The characteristics of included trials comparing Abdominal Paracentesis Drainage with conventional "set-up" approach.

First author (publish year)	Study period	Study type	Sample size		Mean age (years)		Etiology of AP	APACHE Ⅱ at admission		Catheter diameter	Following time	Outcomes[#]
			APD group	Non-APD group	APD group	Non-APD group		APD group	Non-APD group			
Wei-hui Liu[10] (2015)	2009.06-2011.06	RCS	53	49	59±13.5	57±12.7	Gallstone, alcohol abuse, hyperlipemia	15.5±3.55	14.7±3.25	8F-22F	about 20 d from the onset of AP	1,2,3,4,5,6,7
Yin Li[19] (2018)	2012.01-2017.12	RCS	46	42	47.0±14.6	48.8±13.5	NR	NR	NR	8F-16F	in-hospital	1,2,5,6,7
Li-ye Liu[15] (2015)	2013.08-2014.10	PCS	126	129	57±13	55±11	Biliary, alcohol abuse	13.5±3.1	14.2±2.8	8F-22F	in-hospital	1,3,4
Jun-wei Ma[20] (2017)	2009.11-2015.11	RCS	43	39	55.89±4.59	56.28±3.60	Biliary, alcohol abuse	23.69±2.46	22.95±3.26	NR	in-hospital	1,3,4,5,6,7
Zhu Huang[17] (2016)	2010.05-2015.05	RCS	68	64	50 .33±11.17	48.79±9.54	non- HTG induced	15.21 ± 3.48	15.47 ± 3.10	NR	in-hospital	1,6
Wen-ting Luo[18] (2019)	2016.07-2018.11	RCT	49	49	45.69±11.87	44.37± 11.21	NR	17.51±4.47	17.86±4.64	12F	7 day	2,6
Hong-yin Liang[16] (2016)	2015.01-2016.04	RCT	83	78	55.6±14.6	53.7±12.8	Gallstone, alcohol abuse, hyperlipemia	13.4±4.1	13.9±4.9	8F-16F	in-hospital	1,2,5,6
Liang Zhang[21] (2020)	2014.03-2018.11	RCT	84	84	63.22±12.43	63.43±12.37	Biliary, alcohol abuse	NR	NR	NR	NR	2,6,7

APD=Abdominal paracentesis drainage; AP=Acute pancreatitis; NR=Not report; RCS=Retrospective cohort study; PCS=Prospective cohort study; RCT=Randomized control trial; HTG=hypertriglyceridemia

[#]Outcomes: 1=all-cause mortality; 2=the rate of organ failure; 3=the rate of sepsis; 4=the rate of bacteremia; 5=spending; 6=length of hospital stay; 7=the rate of microbial

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infection

Data are mean ± standard deviation unless otherwise indicated

Table 2 The risk of bias assessment of included cohort studies by using the ROBINS-I tool.

Study/Domain	Confounding	Selection of participants into the study	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Bias in selection of the reported result	Overall risk
Wei-hui Liu[10]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
Yin Li[19]	SEVERE	LOW	MODERATE	LOW	LOW	MODERATE	LOW	SEVERE
Li-ye Liu[15]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
Jun-wei Ma[20]	MODERATE	LOW	LOW	LOW	LOW	LOW	LOW	MODERATE
Zhu Huang[17]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW

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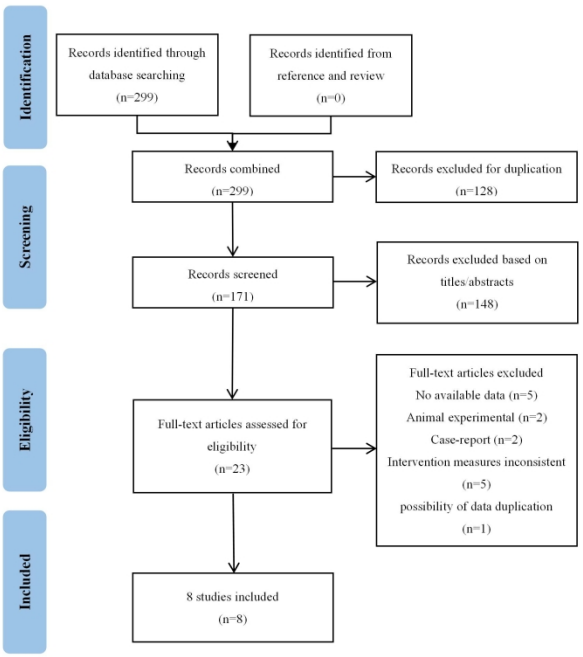


Figure 1 Flow diagram of study selection.

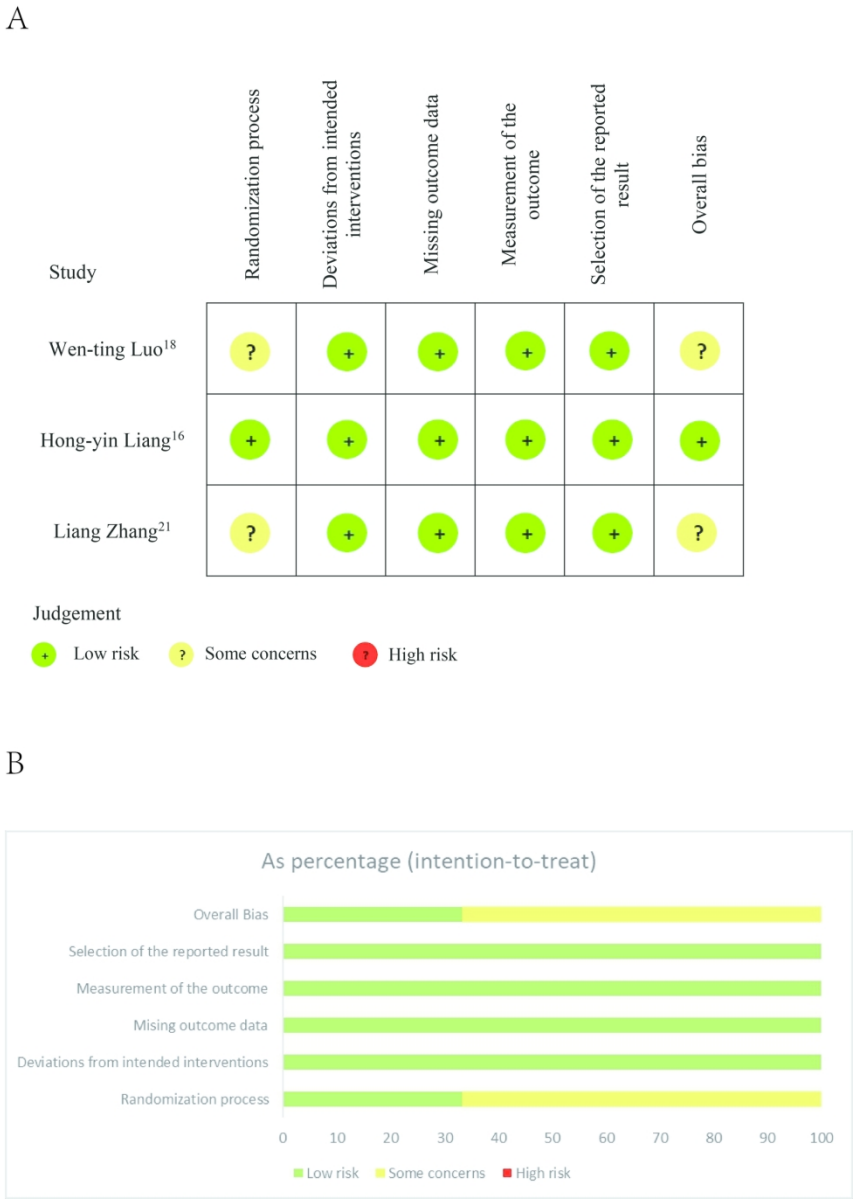


Figure 2: The risk of bias about included RCTs by using ROB-2 tool, A: traffic light plot of RCT bias assessment; B: weighted summary plot of the overall type of bias encountered in RCTs

147x206mm (300 x 300 DPI)

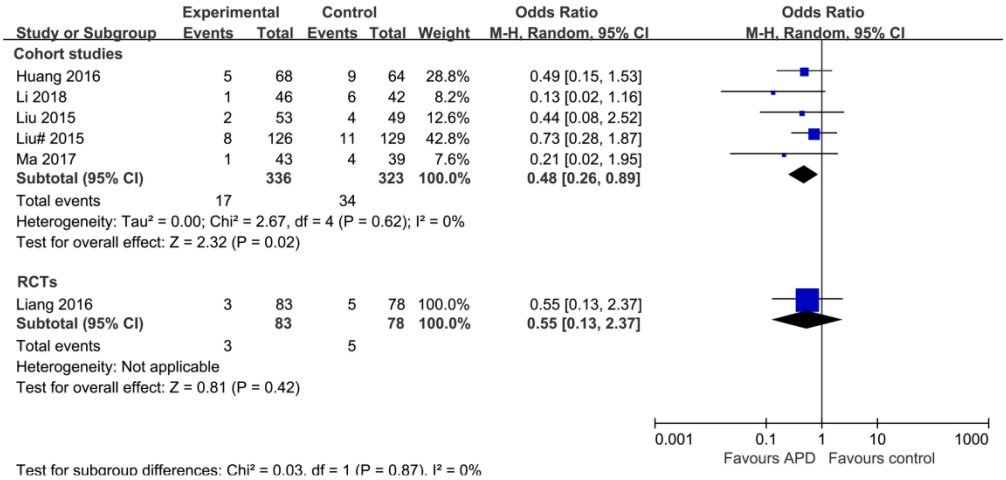


Figure 3: Forest plot for all-cause mortality.

195x93mm (300 x 300 DPI)

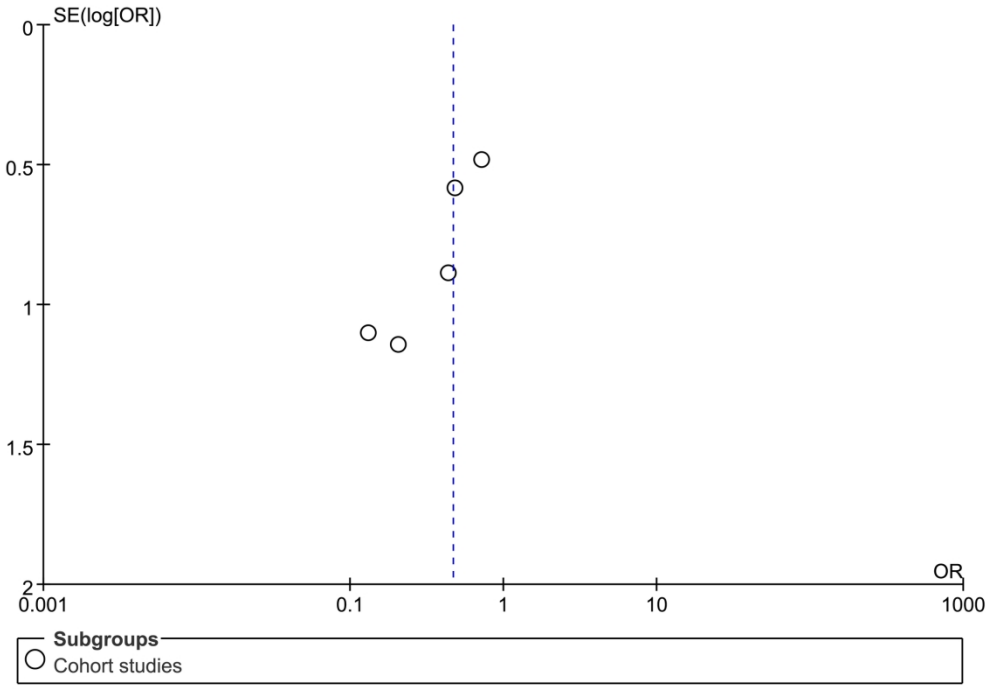


Figure 4: Funnel plot for all-cause mortality.

195x139mm (300 x 300 DPI)

Supplementary Table 1: PubMed search Strategy.

PubMed
#1 “acute pancreatitis” OR “severe acute pancreatitis”
#2 AP OR SAP
#3 Pancreatitis (Mesh)
#4 #1 OR #2 OR #3
#5“Abdominal Paracentesis Drainage” OR APD
#6 “step-up approach”
#7 #5 OR #6
#8 randomized controlled trial
#9 controlled clinical trial
#10 cohort study
#11 RCT OR “observational study” OR “cohort study” OR cohort
#12 #8 OR #9 OR #10 OR #11
#13 #4 AND #7 AND #12

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Supplementary Table 2 GRADE evidence of each predefined outcome.

Outcome	Study type	No. study	Sample size (E/C)	Inconsistency	Indirectness	Imprecision	Large effect	Confounder	Dose-response gradient	GRADE Evidence	Outcome Importance
All-cause Mortality	RCT	1	83/78		Directness	High Imprecise				LOW	Critical
	Cohort study	5	336/323	Consistency	Directness	Imprecise	Large	NO	NO	LOW	
Microbial Infection	RCT	1	84/84		Directness	High Imprecise				Very LOW	Critical
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	LOW	
Sepsis	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Bacteremia	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Organ Failure	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	2	99/91	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	
Length of hospital stay	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	4	210/194	Consistency	Directness	Imprecise	NO	NO	NO	LOW	

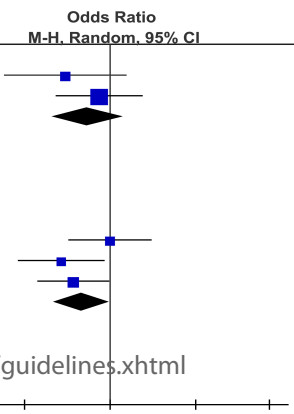
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Spending	RCT	2	167/162	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	

E=experimental group; C=Control group; RCT=Randomized control trial

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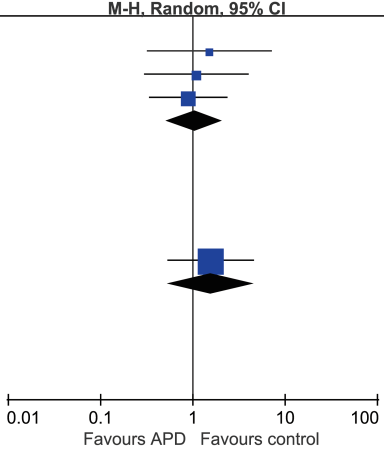
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Cohort studies						
Li 2018	35	46	37	42	33.5%	0.43 [0.14, 1.36]
Liu 2015	17	53	18	49	66.5%	0.81 [0.36, 1.84]
Subtotal (95% CI)		**99**		**91**	**100.0%**	**0.66 [0.34, 1.28]**
Total events	52		55			
Heterogeneity: Tau² = 0.00; Chi² = 0.78, df = 1 (P = 0.38); I² = 0%						
Test for overall effect: Z = 1.23 (P = 0.22)						
RCTs						
Liang 2016	67	83	63	78	31.6%	1.00 [0.46, 2.18]
Liu 2019	17	49	28	49	29.8%	0.40 [0.18, 0.90]
Zhang 2020	19	84	31	84	38.7%	0.50 [0.25, 0.98]
Subtotal (95% CI)		**216**		**211**	**100.0%**	**0.58 [0.35, 0.98]**
Total events	103		122			
Heterogeneity: Tau² = 0.06; Chi² = 2.83, df = 2 (P = 0.24); I² = 29%						
Test for overall effect: Z = 2.05 (P = 0.04)						
Test for subgroup differences: Chi² = 0.08, df = 1 (P = 0.78), I² = 0%						



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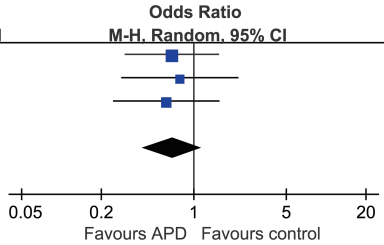
a: microbial infection

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Cohort studies						
Li 2018	43	46	38	42	20.2%	1.51 [0.32, 7.17]
Liu 2015	48	53	44	49	28.8%	1.09 [0.30, 4.02]
Ma 2017	31	43	29	39	51.0%	0.89 [0.33, 2.37]
Subtotal (95% CI)		**142**		**130**	**100.0%**	**1.05 [0.52, 2.12]**
Total events	122		111			
Heterogeneity: Tau² = 0.00; Chi² = 0.32, df = 2 (P = 0.85); I² = 0%						
Test for overall effect: Z = 0.14 (P = 0.89)						
RCTs						
Zhang 2020	78	84	75	84	100.0%	1.56 [0.53, 4.60]
Subtotal (95% CI)		**84**		**84**	**100.0%**	**1.56 [0.53, 4.60]**
Total events	78		75			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.81 (P = 0.42)						
Test for subgroup differences: Chi² = 0.36, df = 1 (P = 0.55), I² = 0%						



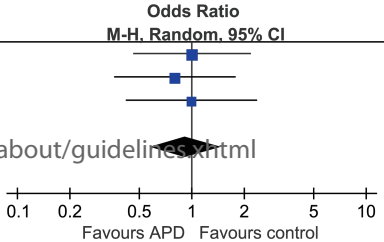
b: sepsis

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Liu 2015	16	53	19	49	41.1%	0.68 [0.30, 1.55]
Liu 2015	7	126	9	129	26.7%	0.78 [0.28, 2.17]
Ma 2017	12	43	15	39	32.2%	0.62 [0.24, 1.57]
Total (95% CI)		**222**		**217**	**100.0%**	**0.69 [0.41, 1.16]**
Total events	35		43			
Heterogeneity: Tau² = 0.00; Chi² = 0.11, df = 2 (P = 0.95); I² = 0%						
Test for overall effect: Z = 1.40 (P = 0.16)						



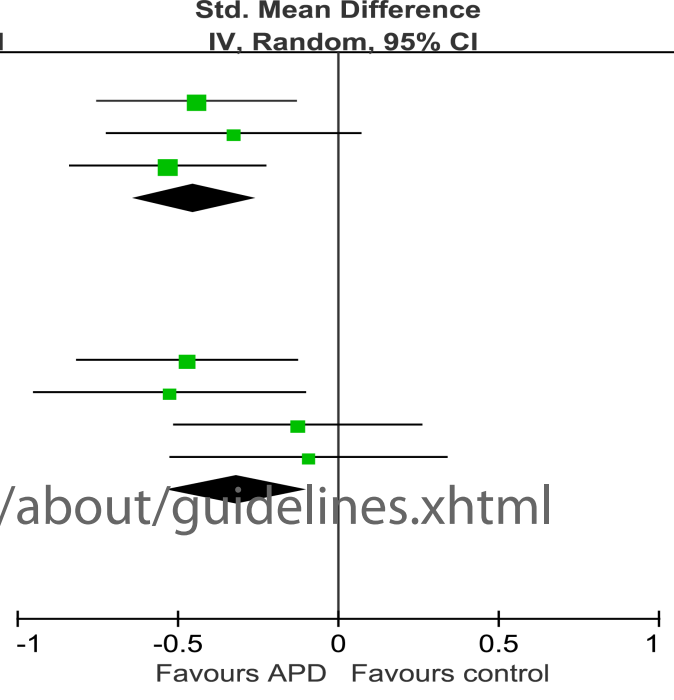
c: bacteremia

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Liu 2015	26	53	24	49	36.5%	1.00 [0.46, 2.18]
Liu 2015	12	126	15	129	34.2%	0.80 [0.36, 1.78]
Ma 2017	22	43	20	39	29.3%	1.00 [0.42, 2.37]
Total (95% CI)		**222**		**217**	**100.0%**	**0.93 [0.58, 1.48]**
Total events	60		59			
Heterogeneity: Tau² = 0.00; Chi² = 0.19, df = 2 (P = 0.91); I² = 0%						
Test for overall effect: Z = 0.32 (P = 0.75)						



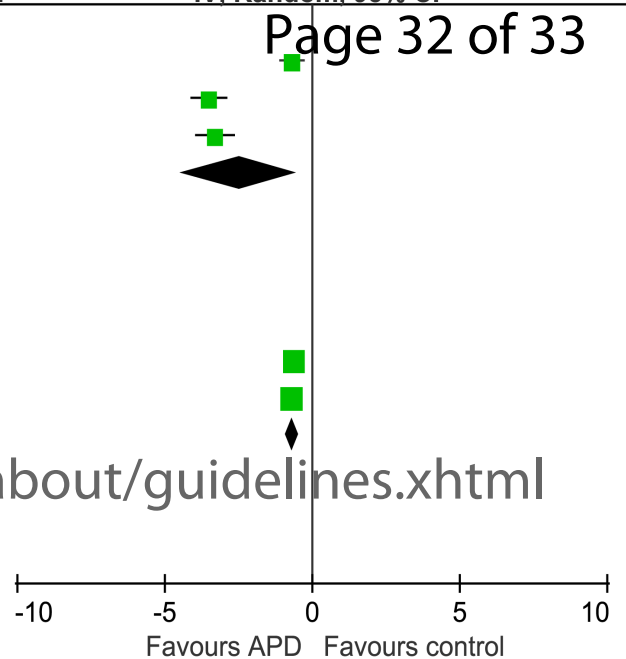
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Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Std. Mean Difference Weight	Std. Mean Difference IV, Random, 95% CI
RCTs								
Liang 2016	28.9	10.7	83	35.3	17.5	78	37.7%	-0.44 [-0.76, -0.13]
Luo 2019	23.16	4.98	49	24.65	4.02	49	23.3%	-0.33 [-0.73, 0.07]
Zhang 2020	47.92	9.28	84	52.63	8.31	84	39.0%	-0.53 [-0.84, -0.22]
Subtotal (95% CI)			**216**			**211**	**100.0%**	**-0.45 [-0.64, -0.26]**
Heterogeneity: Tau² = 0.00; Chi² = 0.64, df = 2 (P = 0.72); I² = 0%								
Test for overall effect: Z = 4.59 (P < 0.00001)								
Cohort studies								
Huang 2016	52.78	22.51	68	66.13	33.07	64	30.9%	-0.47 [-0.82, -0.13]
Li 2018	47.93	9.27	46	52.62	8.32	42	22.1%	-0.53 [-0.95, -0.10]
Liu 2015	58.9	31.24	53	63.2	36.15	49	25.7%	-0.13 [-0.52, 0.26]
Ma 2017	55.89	4.59	43	56.28	3.6	39	21.4%	-0.09 [-0.53, 0.34]
Subtotal (95% CI)			**210**			**194**	**100.0%**	**-0.31 [-0.53, -0.10]**
Heterogeneity: Tau² = 0.01; Chi² = 3.64, df = 3 (P = 0.30); I² = 18%								
Test for overall effect: Z = 2.83 (P = 0.005)								
Test for subgroup differences: Chi² = 0.84, df = 1 (P = 0.36), I² = 0%								



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Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Std. Mean Difference Weight	Std. Mean Difference IV, Random, 95% CI
Cohort studies								
Li 2018	46,541.93	8,326.67	46	52,518.43	8,874.6	42	33.8%	-0.69 [-1.12, -0.26]
Liu 2015	46,152.2	2,475.13	53	60,013.1	5,041.02	49	33.2%	-3.51 [-4.13, -2.88]
Ma 2017	46,200	2,600	43	60,100	5,400	39	33.0%	-3.30 [-3.98, -2.63]
Subtotal (95% CI)			**142**			**130**	**100.0%**	**-2.49 [-4.46, -0.51]**
Heterogeneity: Tau² = 2.96; Chi² = 71.99, df = 2 (P < 0.00001); I² = 97%								
Test for overall effect: Z = 2.47 (P = 0.01)								
RCTs								
Liang 2016	8,174.1	2,075.8	83	10,572.1	5,041.3	78	49.2%	-0.63 [-0.94, -0.31]
Zhang 2020	46,500	8,400	84	52,600	8,800	84	50.8%	-0.71 [-1.02, -0.39]
Subtotal (95% CI)			**167**			**162**	**100.0%**	**-0.67 [-0.89, -0.44]**
Heterogeneity: Tau² = 0.00; Chi² = 0.12, df = 1 (P = 0.73); I² = 0%								
Test for overall effect: Z = 5.88 (P < 0.00001)								
Test for subgroup differences: Chi² = 3.22, df = 1 (P = 0.07), I² = 68.9%								



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6-7
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12-15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis

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Primary Subject Heading:	Intensive care
Secondary Subject Heading:	Gastroenterology and hepatology, Emergency medicine
Keywords:	Pancreatic disease < GASTROENTEROLOGY, Adult intensive & critical care < INTENSIVE & CRITICAL CARE, Minimally invasive surgery < GYNAECOLOGY

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Efficacy and Safety of Abdominal Paracentesis Drainage on Acute Pancreatitis Patients: A Systematic Review and Meta-Analysis

Zongqing Lu[1,2] , Xingxing Zhu[1,2], Tian-feng Hua[1,2], Jin Zhang[1,2], Wenyan Xiao[1,2], Di Jia[1,2], Min Yang[1,2*]

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Conflicts of Interest

The authors declare that there is no conflict of interest.

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Abbreviations

AP: Acute pancreatitis

APD: Abdominal paracentesis drainage

PCD: Percutaneous catheter drainage

MAP: Mild acute pancreatitis

MSAP: Moderate severe acute pancreatitis

SAP: Severe acute pancreatitis

RIS: Required information size

CLS: Capillary leak syndrome

IAP: Intra-abdominal pressure

RCT: Randomized controlled trial

OR: Odds ratio

CI: Confidence interval

SMD: Standardized mean difference

CNKI: China National Knowledge Infrastructure

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Abstract

Objectives

To conduct a systematic review and meta-analysis of the efficacy and safety of abdominal paracentesis drainage (APD) in acute pancreatitis (AP) patients when compared with conventional "step-up" strategy based on percutaneous catheter drainage (PCD).

Design

Systematic review and meta-analysis.

Methods

PubMed, EMBASE, Cochrane Library, MEDLINE (OVID), CNKI and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) from inception to July 25, 2020. Studies related to comparing APD with conventional "step-up" strategy based on PCD were included.

Outcomes

The primary outcome was all-cause mortality. The secondary outcomes were the rate of organ dysfunction, infectious complications, hospitalization expenses and length of hospital stay.

Results

Five cohort studies and three RCTs were included in the analysis. Compared with the conventional "step-up" method, pooled results suggested APD significantly decreased all-cause mortality during hospitalization (Cohort studies: OR 0.48, 95%CI 0.26-0.89, P=0.02), length of hospital stay (Cohort studies: SMD -0.31, 95%CI -0.53, -0.10, P=0.005; RCTs: SMD -0.45, 95%CI -0.64, -0.26, P<0.001) and hospitalization expenses (Cohort studies: SMD -2.49, 95%CI -4.46, -0.51, P<0.001; RCTs: SMD -0.67, 95%CI -0.89, -0.44, P<0.001). There was no evidence to prove that APD was associated with a higher incidence of infectious complications. However, the

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incidence of organ dysfunction between cohort studies and RCTs subgroup slightly differed (Cohort studies: OR 0.66, 95%CI 0.34-1.28, P=0.22; RCTs: OR 0.58, 95%CI 0.35-0.98, P=0.04).

Conclusions

The findings suggest that early application of APD in AP patients is associated with reduced all-cause mortality, expenses during hospitalization and the length of stay compared with the "step-up" strategy without significantly increased the risk of infectious complications. These results must be interpreted with caution because of the limited number of included studies as well as a larger dependence on observational trials.

KEYWORDS: abdominal paracentesis drainage; acute pancreatitis; percutaneous catheter drainage; systematic review; meta-analysis

Strengths and limitations of this study

1. This trial was the first systematic review and meta-analysis comparing abdominal paracentesis drainage (APD) with conventional percutaneous catheter drainage (PCD) method in treating acute pancreatitis (AP) patients.

2. Thorough searching of six major electronic databases, including two major non-English language databases.

3. The study protocol has been registered in PROSPERO and reported according to PRISMA guidelines.

4. This study relied greatly on cohort studies due to the limited number of published RCTs, which means that final pooled results must be interpreted very cautiously.

5. The meta-analysis results might be affected by the length of follow-up, canal diameter of APD and etiology given the limited numbers of studies. Confirmation of findings by further clinical trials is warranted.

1. Introduction

Acute pancreatitis (AP), a rapidly progressing disease characterized by severe complications and high mortality, leads to many admissions to intensive care units [1]. According to the updated 2012 Atlanta Classification, AP is

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classified into mild acute pancreatitis (MAP), moderate-severe acute pancreatitis (MSAP), and severe acute pancreatitis (SAP). MAP particularly causes upper abdominal pain, moreover, MSAP and SAP are associated with local or systemic complications, including peripancreatic fluid collection, which causes persistent organ dysfunction eventually [2].

Early conservative treatments of AP involve active rehydration, retropyloric feeding and pancreatin inhibitors. Once abdominal compartment syndrome and/or intestinal ischemia occur, surgery is usually required to remove the necrotic tissue [3]. However, it had been reported that open surgery is connected to high mortality, infectious complications and prolonged hospitalization [4]. *Freeny et al* (1998), for the first time, reported a novel treatment, commonly known as the "step-up" method based on percutaneous catheter drainage (PCD), whose purpose was to progressively manage infections rather than immediately eliminates necrosis [5]. The efficacy and safety of the "step-up" method had confirmed through further clinical trials [6,7]. Although most acute peripancreatic fluid collection and acute necrotic collection were detected from week 2 to week 3 after AP onset, PCD was suggested to be performed at least 4 weeks later until the necrosis formed a wrap [8,9]. It indicated that PCD seemed not to be optimum as an early invasive intervention and management. Recently, a retrospective cohort study to this problem had reported a new insert catheters technique, be called as the abdominal paracentesis drainage (APD). It was performed via the right paracolic sulci or left paracolic sulci at abdominal collection volume was greater than 50ml [10]. The results demonstrated that early APD treatment could effectively lessen the release of inflammatory factors and improve the clinical prognosis. However, it is uncertain that the use of APD is associated with an increased risk of exogenous infection in AP patients.

Considering the current controversies, the purpose of this systematic review was to compare the efficacy and safety of APD with the "step-up" strategy in published clinical trials of patients AP patients.

2. Material and Methods

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The present systematic review and meta-analysis were reported in accordance with the Preferred Reporting item for Systematic Review and Meta-analysis (PRISMA) statement [11]. And, this review has been registered on the PROSPERO website as No. CRD42020168537.

2.1 Data sources and searches

PubMed, EMBASE, Cochrane Library, MEDLINE (OVID), China National Knowledge Infrastructure (CNKI) and Wanfang Database were electronically searched to collect cohort studies and randomized controlled trials (RCTs) about the APD for the treatment of AP from inception to February 15, 2020, by using keywords combined with Medical Subject Headings (MeSH) terms, regardless of language and region. We retrieved it again on July 25, 2020, to included the latest published paper. In addition, the reference lists of related literature were manually searched for possible trials. The search strategy for PubMed was shown in Supplementary Table 1.

2.2 Study selection

Zongqing Lu and Xingxing Zhu searched independently, according to predefined inclusion and exclusion criteria. Duplicate literature deletion, title and abstract screening for relevance had been done using Endnote software. Then, the full text was acquired to determine inclusion eligibility eventually. Any disagreement was resolved through discussion, a third review author (Min Yang) participated where necessary.

Published RCTs and cohort studies meeting the following criteria were included: 1.population: Adult patients with acute pancreatitis regardless of pathogenesis, however, detailed diagnostic criteria for acute pancreatitis must be required, and each participant must conform to the requirement of APD: 1) enough volume of celiac fluid collections (> 50mL); 2) a feasible pathway existed. No limitation on race and nationality. 2.intervention: APD was administered within 4 weeks of onset. 3.comparison: traditional “step-up strategy” in which APD was not performed. 4.the sample size of each group exceeds 20.

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2.3 Outcomes and Data Extraction

Two authors (Zongqing Lu and Di Jia) independently extracted data using a pre-piloted form designed by Excel 2019 software (Microsoft Corporation), and the results were confirmed by another author (Tianfeng Hua). The collected data include the first author, publish year, sample size, mean age and sex ratio of each group, study period, following time, cause of AP, APACHE Ⅱ scores at admission, catheter diameter and outcomes data. If any information above was inadequate, we contacted the original author via email to consult related data. We resolved discrepancies through discussion. The predefined primary outcome was all-cause mortality during hospitalization. The secondary outcomes were the rate of organ dysfunction, the rate of related infectious complications such as microbial infection, sepsis and bacteremia after treatment, expenses during hospitalization and the length of hospital stay.

2.4 Quality Assessment

The quality of filtered articles has been assessed by two authors (Wenyan Xiao and Jin Zhang) respectively. The risk of bias tool performed was specific to the study type. For RCTs, the revised Cochrane tool to assess risk of bias (ROB-2) was used for each of the following five domains: randomization process, deviations from intended interventions, missing outcomes data, measurement of the outcome and selection of the reported results [12]. We reviewed each RCT and classifies them as low, probably low, probably high or high risk of bias. However, blinding of patients and clinicians was often unrealistic in invasive treatment, and which has limited influence on primary outcomes. So, trials were seen as low risk in the domain of deviations from intended interventions and measurement of the outcome even without double-blinding.

Meanwhile, cohort studies were assessed using the Risk Of Bias In Non-randomised Studies - of Interventions tool (ROBINS-I), in which seven key domains were contained: 1.confounding; 2.selection of participants; 3.classification of interventions; 4.deviations from intended interventions; 5.missing data; 6.measurement of

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outcomes; 7.selection of the reported results It was widely used in the non-RCT evaluation, and recommended by the Cochrane collaboration [13].

2.5 Quality of evidence

Two authors assessed the quality of each evidence respectively by using the GRADE system (Grading of Recommendations Assessment, Development and Evaluation) for risk of bias, inconsistency, indirectness, imprecision, publication bias, large effect, confounder and dose-response gradient [14]. The quality was divided into very low, low, moderate, or high. The results were generated by using the GRADE Profiler.

2.6 Statistical Analysis

RevMan 5.3 (Nordic Cochrane Centre) was applied for statistical analysis of the included studies. The odds ratio (OR) and the 95% confidence intervals (CIs) were calculated for dichotomous data with the Mantel-Haenszel method. For continuous data, we calculated the standard mean difference (SMD) and 95%CIs, because of the differences in measurement methods and units. *P* values less than 0.05 denoted statistical significance. Meta-analysis was performed if there were predefined outcomes from two or more studies. Statistical heterogeneity across trials was examined using the I^2 statistical tests and *P* values. Values with $P<0.1$ and I^2 greater than 50% denoted significant heterogeneity. the statistical test for heterogeneity may not be sensitive due to the limited number of component studies and patients. We employed the random-effects model to merge each result because the fix-effects model could amplify the weight of large sample studies. Subsequently, subgroup analysis according to the RCTs or cohort studies was performed when sufficient data was available to avoid the influence of inappropriate pooled results on the validity of evidence. Sensitivity analysis was carried out by the leave-one-out method to evaluate the sources of heterogeneity and the feasibility of results. The funnel plot and Egger's test were performed to detect potential publication bias.

3. Results

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3.1 Literature Research

The flow diagram showed the process of literature screening, selection and reasons for exclusion (Fig.1). Our initial search yielded 299 records. After removing duplication and reviewing the titles/abstracts, 23 articles were considered to be potentially eligible for inclusion. After reading the full text, 15 studies were excluded for the following reasons: no available data (n=5), animal experimental (n=2), case-report (n=2), intervention measures inconsistent (n=5) and the possibility of data duplication (n=1). As a result, eight studies [10,15-21] were eventually included in this meta-analysis.

3.2 Trials Characteristics

The characteristics of the eligible trials had been summarized in Table 1. There were five cohort studies and three RCTs published between 2014 and 2020, which compared the clinical prognosis of AP patients who were cured by APD or conventional "step-up" approach. Population sizes ranged from 82 to 255, with a total of 1086 patients. Except for the catheter diameter and the catheter numbers, there was no statistical difference of baseline between APD and control groups in each trial, according to the report.

3.3 Risk of Bias Assessment and GRADE Profile Evidence

The details about the risk of bias were respectively shown in Table 2 and Figure 2. For included cohort studies, three studies were classified as low risk of bias [10, 15, 17], *Li et al* were categorized as a severe risk of bias due to the lack of necessary baseline comparison and we could not determine whether the outcome indicators were measured at the same time of each group in this trial[19]. Meanwhile, *Ma et al* did not report an appropriate method to balance the treatment start time, so we categorize it as moderate risk through discussion [20]. For RCTs included, one was categorized as low risk of bias in each domain [16], others were considered as some concerns because they did not report the way of allocation concealment [18,21].

GRADE evidence profiles are showed in Supplementary Table 2. Due to either unclear risk of bias or

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imprecision resulted from small sample size influence and relative wide confidence interval, the quality of evidence of outcomes in the RCTs subgroup was low or very low, according to the GRADE framework. In addition, we also determined the GRADE evidence of cohort studies subgroup as low or very low eventually, considering the natural risk of observational trials, and small effect size except for all-cause mortality.

3.4 Primary outcome

All-cause mortality during hospitalization was reported in six trials, which included 820 patients [10,15-17,19,20]. Compared with the conventional "step-up" approach, five cohort studies [10,15,17,19,20] of APD intervention reported a significantly reduced risk of all-cause mortality in acute pancreatitis patients (OR 0.48, 95%CI 0.26-0.89, P=0.02) with low heterogeneity (I^2=0.0%) (Fig.3). In RCTs subgroup, there was no statistical differences in all-cause mortality (n=161, OR 0.55, 99%CI 0.13-2.37, P=0.42) [16].

3.5 Secondary outcomes:

3.5.1 The rate of organ dysfunction: Five studies provided data on the rate of organ dysfunction with a total of 617 patients [10,16,18,19,21]. Compared with the conventional "step-up" approach, the results of cohort studies subgroup [10,19] shown that there were no significant differences in the incidence of organ dysfunction (n=190, OR 0.66, 95%CI 0.34-1.28, P=0.22). However, the results of RCTs subgroup [16,18,21] supported APD (n=427, OR 0.58, 95%CI 0.38-0.98, P=0.04) (Supplementary Figure 1). We considered that the inconsistencies of results between the two subgroups resulted from the difference in sample size because preferred APD trends have appeared in cohort studies subgroup. The heterogeneity in cohort studies and RCTs subgroup was I^2=0% and I^2=29%, respectively.

3.5.2 Infectious complications: Five studies reported the number of infectious complications, which included microbial infection [10,19-21], sepsis [10,15,20] and bacteremia [10,15,20], with a total of 695 patients. We pooled data from three cohort studies[10,19,20]which showed no significant differences in the incidence of microbial infection

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(OR 1.05, 95%CI 0.52-2.12, *P*=0.89) with I^2=0%, and this concurred with another RCT's result (OR 1.56, 95%CI 0.53-4.60, *P*=0.42)[21]. For the incidence of sepsis and bacteremia, the pooled results from three cohort studies showed that APD could not significantly increase the related risk versus the conventional "step-up" approach (sepsis: OR 0.69, 95%CI 0.41-1.16, *P*=0.16; bacteremia: OR 0.93, 95%CI 0.58-1.48, *P*=0.75). The heterogeneity in both of them was I^2=0%. (Supplementary Figure 2)

3.5.3 Length of hospital stay: Four cohort studies (including 404 patients) [10,17,19,20] and three RCTs (including 427 patients) [16,18,21] were included in this analysis. The results of the cohort studies subgroup revealed that APD significantly decreased the length of hospital stay when compared with the conventional "step-up" approach (SMD -0.31, 95%CI -0.53, -0.10, *P*=0.005). This result was also consistent with that for the RCTs subgroup. (SMD -0.45, 95%CI -0.64, -0.26, *P*<0.001). And the heterogeneity in each subgroup was I^2=18% and I^2=0%. (Supplementary Figure 3).

3.5.4 Hospitalization expenses: We extracted the data about spending during hospitalization from three cohort studies and two RCTs, with a total of 601 patients [10,16,19-21]. The pooled results of cohort studies subgroup [10,19,20] demonstrated that APD could significantly reduce the spending versus conventional "step-up" approach (SMD -2.49, 95%CI -4.46, -0.51, *P*<0.001), which was consistent with the result of RCTs subgroup [16,21] (SMD -0.67, 95%CI -0.89, -0.44, *P*<0.001), and heterogeneity were I^2=97%, I^2=0% in cohort studies and RCTs subgroup respectively. (Supplementary Figure 4)

3.6 Publication Bias and Sensitivity Analysis

We just included cohort studies to explore the potential publication bias for primary outcomes, due to the limited number of RCT. The funnel plot demonstrated an unsymmetrical shape (Fig.4). In addition, the subsequent Egger's test shown *P*<.05 (*P*=0.025), which suggested a possible publication bias exists. Sensitivity analysis was conducted by the leave-one-out method. For the spending, we found that the I^2 value decreased to 0% after

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excluding the trial conducted by *Li et al* (2018), in which the retroperitoneal catheter drainage was performed [19];. however, the pooled result was stable by using sensitivity analysis. Finally, we excluded studies with a severe risk of bias and finally proved that the pooled results were still stable. Thus, the high heterogeneity may result from the difference in sample size, the quality of trial and PCD pathways.

4. Discussion

The present systematic review and meta-analysis of three RCTs and five cohort studies demonstrated that APD is an effective and safe intervention for the treatment of AP. In line with evidence based on cohort studies, we found that APD had significant decreased trends toward all-cause mortality, length of stay and expenses as compared to the conventional "step-up" treatment, that is consistent with the findings of high-quality evidence from the RCTs, but no conspicuous difference in the risk of extra infection and the rate of organ dysfunction.

Pancreatic ascites was enriched in proteases, along with lipases and cytokines, which would be a natural medium for specific intestinal bacteria [22]. Several factors could account for its formation: (i) Pancreatic local inflammation activated host cytokine in the first to the second week [2]. Excessive inflammatory cascades aggravated pancreatic vasospasm and capillary permeability, causing massive plasma extravasation and extensive cell necrosis. (ii) A large amount of intravenous fluid infusion and capillary leak syndrome (CLS) could exacerbate the accumulation of peritoneal fluid. Thus, it is important to inhibit the activation and release of those inflammatory cytokines. Peritoneal lavage, within 48 hours after SAP, could decrease serum levels of TNF-α and IL-6, and increase IL-10 levels [23], but remained controversial in short-term mortality and complications [24,25,26]. Instead, the protective effects of PCD on the outcomes of AP patients had been demonstrated, but it is not suitable for early intervention [27]. Compared to the above-mentioned two methods, APD could not only significantly decrease serum IL-6, TNF-α, C-reaction protein levels and the incidence of

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subsequent minimally invasive necrosectomy, but also improve survival rates [10,16,20]. Furthermore, it was suggested to operate as soon as possible if abdominal or pelvic cavity fluid accumulated more than 50 mL.

Furthermore, possibly owing to the relief of intra-abdominal pressure (IAP), APD is more effective than the conventional "step-up" strategy. Intra-abdominal hypertension (IAH) has been established as relevant for the inflammatory response, CLS-induced abdominal effusion, and visceral edema [28]. IAH damaged the intestinal mucosal perfusion, so as the intestinal barrier, which consequently increases the possibility of bacterial translocation. It also had some adverse effects on the host circulatory system, which was mirrored by the reduction of venous return and cardiac output.

As an invasive intervention, the operator routinely focuses on the additional risks of infectious complications because the impact of exogenous infection has a fatal impact on critical patients and can offset the endeavor by previous treatment. This systematic review concluded that APD did not increase the risk of microbial infection, sepsis and bacteremia as compared with PCD. Similar results were reported in trials with a low risk of bias.

Some limitations should be acknowledged. Firstly, this review relied greatly on cohort studies because published RCTs are limited, and it is generally known that observational studies have the risk of selection bias. Secondly, although sensitivity analysis proved the stability of results of the expenses, the heterogeneity was very large; as such, we needed to be cautious about this result. Thirdly, the effect of APD could be influenced by the difference in catheter position, catheter lumen, catheter time, and diameter. Fourthly, the morbidity of organ dysfunction and related infectious complications are likely to be influenced by the length of follow-up time. Nonetheless, there were insufficient subgroups to identify this. Lastly, the sample size in the included studies was small, and in the meta-analysis, the small study effect in small trials could magnify the positive effects of the intervention compared with large trials [29]. The difference in the results of the cohort studies and the RCTs subgroups could also be explained partly by the fact that component studies were less and with a small sample size. Although the

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search strategy has been ameliorated and strictly followed the PRISMA statement, we should be careful when interpreting the final pooled results.

Because the role of APD in improving the prognosis of patients with AP remains unclear, there are currently no guidelines or consensus for APD. In the future, researchers and clinicians should focus on the following questions: Firstly, which subgroup of AP can benefit from APD? Secondly, when is the optimal timing to initiate APD? Finally, is the combination of APD and dynamic monitoring of intra-abdominal pressure more beneficial for improving the prognosis of AP patients?

5. Conclusions

The findings suggest that early application of APD compared with "step-up" strategy is significantly associated with reduced all-cause mortality, expenses during hospitalization and the length of stay in AP patients, without increased infectious complications. These results must be interpreted with caution because of the limited number of included studies as well as a larger dependence on observational trials.

Author contributions

Zongqing Lu and Xingxing Zhu contributed equally to this work, who designed the study, collected and analyzed data, drafted and revised the article, and final approved the version to be published; Min Yang and Tian-feng Hua contributed to the interpretation of data, revising of the article and final approval of the version to be published; Wenyan Xiao, Jin Zhang and Di Jia contributed to acquisition of data, analysis data, drafting the article.

Data availability statement

Data are available upon reasonable request. All data relevant to the study are included in the article or uploaded as supplementary information. The data set used and analyzed for this review will be available from the

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corresponding author upon a reasonable request.

Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Ethics Statement

This study did not require ethical approval as the data used have been published previously, and hence are already in the public domain.

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Figure Legends:

Figure 1: Flow diagram of study selection.

Figure 2: The risk of bias about included RCTs by using ROB-2 tool, A: traffic light plot of RCT bias assessment; B: weighted summary plot of the overall type of bias encountered in RCTs

Figure 3: Forest plot for all-cause mortality.

Figure 4: Funnel plot for all-cause mortality.

Supplementary Figure 1: Forest plot for the rate of organ dysfunction.

Supplementary Figure 2: Forest plot for the risk of infectious complications.

Supplementary Figure 3: Forest plot for the length of hospital stay.

Supplementary Figure 4: Forest plot for hospitalization expenses.

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Table 1 The characteristics of included trials comparing Abdominal Paracentesis Drainage with conventional "step-up" approach.

First author (publish year)	Study period	Study type	Sample size		Mean age (years)		Etiology of AP	APACHE Ⅱ at admission		Catheter diameter	Following time	Outcomes[#]
			APD group	Non-APD group	APD group	Non-APD group		APD group	Non-APD group			
Wei-hui Liu[10] (2015)	2009.06-2011.06	RCS	53	49	59±13.5	57±12.7	Gallstone, alcohol abuse, hyperlipemia	15.5±3.55	14.7±3.25	8F-22F	about 20 d from the onset of AP	1,2,3,4,5,6,7
Yin Li[19] (2018)	2012.01-2017.12	RCS	46	42	47.0±14.6	48.8±13.5	NR	NR	NR	8F-16F	in-hospital	1,2,5,6,7
Li-ye Liu[15] (2015)	2013.08-2014.10	PCS	126	129	57±13	55±11	Biliary, alcohol abuse	13.5±3.1	14.2±2.8	8F-22F	in-hospital	1,3,4
Jun-wei Ma[20] (2017)	2009.11-2015.11	RCS	43	39	55.89±4.59	56.28±3.60	Biliary, alcohol abuse	23.69±2.46	22.95±3.26	NR	in-hospital	1,3,4,5,6,7
Zhu Huang[17] (2016)	2010.05-2015.05	RCS	68	64	50 .33±11.17	48.79±9.54	non- HTG induced	15.21 ± 3.48	15.47 ± 3.10	NR	in-hospital	1,6
Wen-ting Luo[18] (2019)	2016.07-2018.11	RCT	49	49	45.69±11.87	44.37± 11.21	NR	17.51±4.47	17.86±4.64	12F	7 day	2,6
Hong-yin Liang[16] (2016)	2015.01-2016.04	RCT	83	78	55.6±14.6	53.7±12.8	Gallstone, alcohol abuse, hyperlipemia	13.4±4.1	13.9±4.9	8F-16F	in-hospital	1,2,5,6
Liang Zhang[21] (2020)	2014.03-2018.11	RCT	84	84	63.22±12.43	63.43±12.37	Biliary, alcohol abuse	NR	NR	NR	NR	2,6,7

APD=Abdominal paracentesis drainage; AP=Acute pancreatitis; NR=Not report; RCS=Retrospective cohort study; PCS=Prospective cohort study; RCT=Randomized control trial; HTG=hypertriglyceridemia.

[#]Outcomes: 1=all-cause mortality; 2=the rate of organ dysfunction; 3=the rate of sepsis; 4=the rate of bacteremia; 5=hospitalization expenses; 6=length of hospital stay; 7=the

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rate of microbial infection.

Data are mean ± standard deviation unless otherwise indicated.

Table 2 The risk of bias assessment of included cohort studies by using the ROBINS-I tool.

Study/Domain	Confounding	Selection of participants into the study	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Bias in selection of the reported result	Overall risk
Wei-hui Liu[10]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
Yin Li[19]	SEVERE	LOW	MODERATE	LOW	LOW	MODERATE	LOW	SEVERE
Li-ye Liu[15]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
Jun-wei Ma[20]	MODERATE	LOW	LOW	LOW	LOW	LOW	LOW	MODERATE
Zhu Huang[17]	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW

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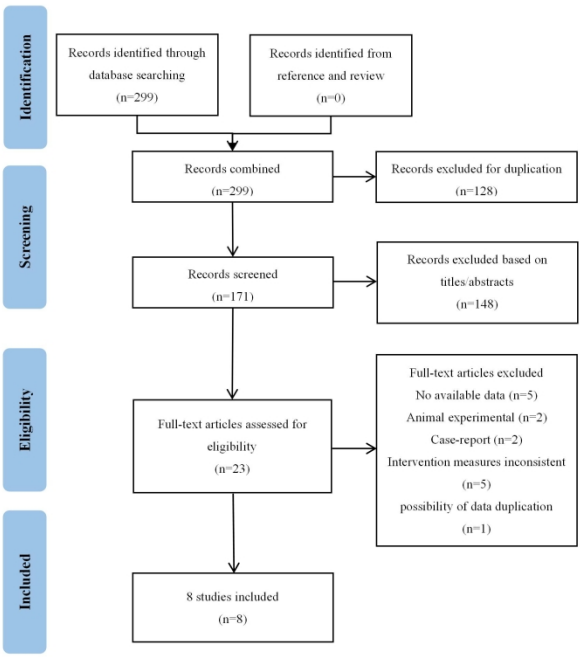


Figure 1 Flow diagram of study selection.



Figure 2: The risk of bias about included RCTs by using ROB-2 tool, A: traffic light plot of RCT bias assessment; B: weighted summary plot of the overall type of bias encountered in RCTs

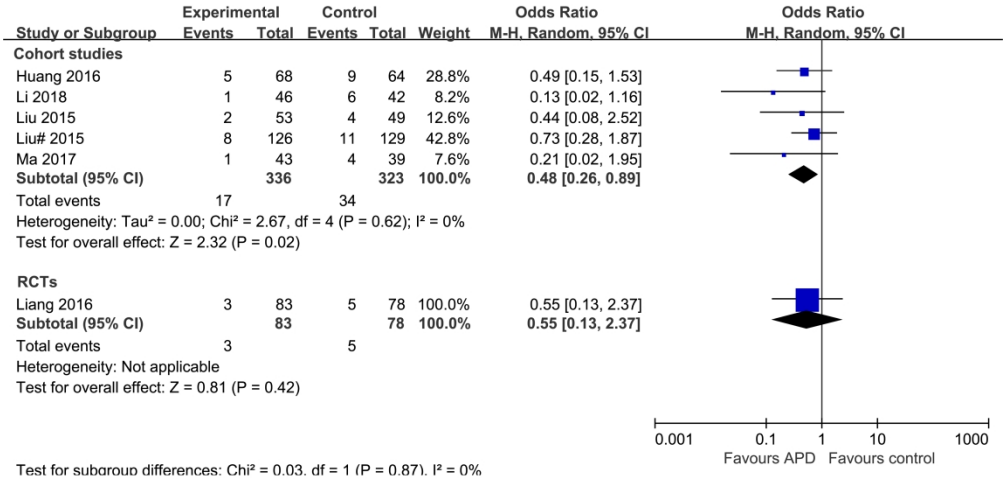


Figure 3: Forest plot for all-cause mortality.

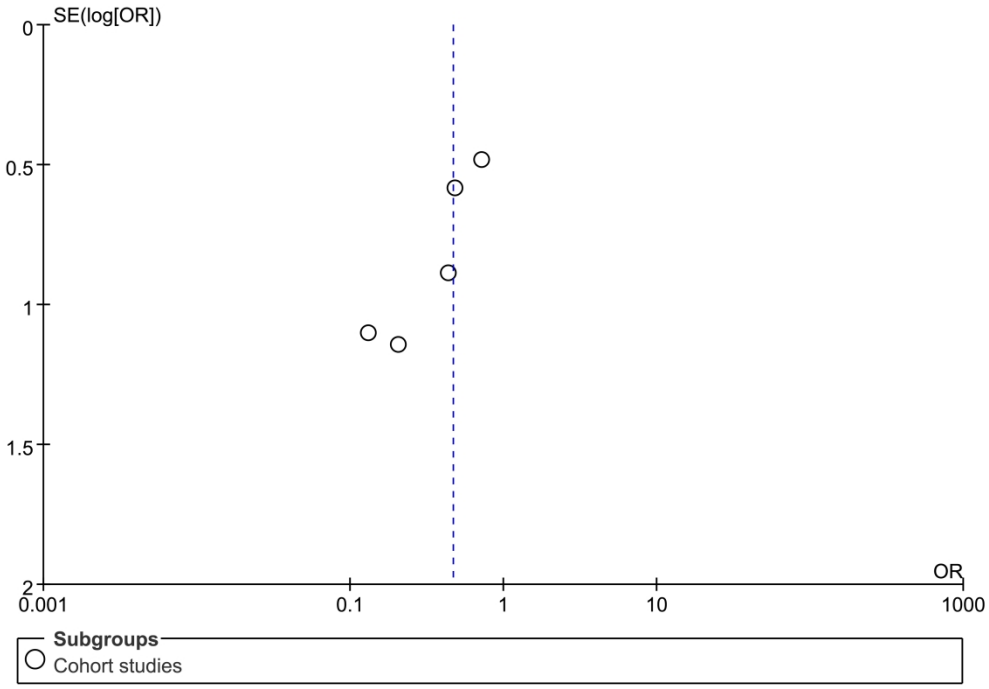


Figure 4: Funnel plot for all-cause mortality.

Supplementary Table 1: PubMed search Strategy.

PubMed
#1 “acute pancreatitis” OR “severe acute pancreatitis”
#2 AP OR SAP
#3 Pancreatitis (Mesh)
#4 #1 OR #2 OR #3
#5“Abdominal Paracentesis Drainage” OR APD
#6 “step-up approach”
#7 #5 OR #6
#8 randomized controlled trial
#9 controlled clinical trial
#10 cohort study
#11 RCT OR “observational study” OR “cohort study” OR cohort
#12 #8 OR #9 OR #10 OR #11
#13 #4 AND #7 AND #12

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Supplementary Table 2 GRADE evidence of each predefined outcome.

Outcome	Study type	No. study	Sample size (E/C)	Inconsistency	Indirectness	Imprecision	Large effect	Confounder	Dose-response gradient	GRADE Evidence	Outcome Importance
All-cause Mortality	RCT	1	83/78		Directness	High Imprecise				LOW	Critical
	Cohort study	5	336/323	Consistency	Directness	Imprecise	Large	NO	NO	LOW	
Microbial Infection	RCT	1	84/84		Directness	High Imprecise				Very LOW	Critical
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	LOW	
Sepsis	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Bacteremia	Cohort study	3	222/217	Consistency	Directness	Imprecise	NO	NO	NO	LOW	Critical
Organ Failure	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	2	99/91	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	
Length of hospital stay	RCT	3	216/211	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	4	210/194	Consistency	Directness	Imprecise	NO	NO	NO	LOW	

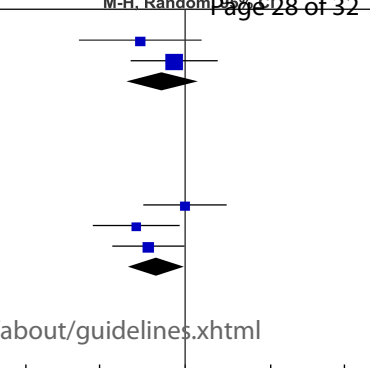
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Spending	RCT	2	167/162	Consistency	Directness	Imprecise				LOW	Important
	Cohort study	3	142/130	Consistency	Directness	Imprecise	NO	NO	NO	Very LOW	

E=experimental group; C=Control group; RCT=Randomized control trial

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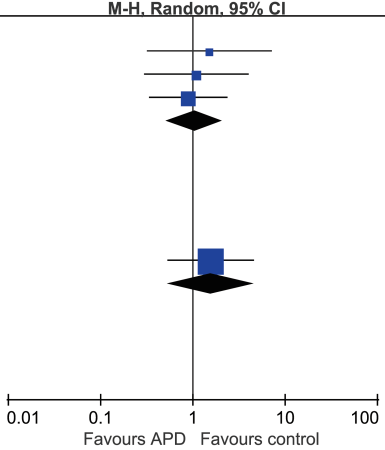
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Cohort studies						
Li 2018	35	46	37	42	33.5%	0.43 [0.14, 1.36]
Liu 2015	17	53	18	49	66.5%	0.81 [0.36, 1.84]
Subtotal (95% CI)		**99**		**91**	**100.0%**	**0.66 [0.34, 1.28]**
Total events	52		55			
Heterogeneity: Tau² = 0.00; Chi² = 0.78, df = 1 (P = 0.38); I² = 0%						
Test for overall effect: Z = 1.23 (P = 0.22)						
RCTs						
Liang 2016	67	83	63	78	31.6%	1.00 [0.46, 2.18]
Liu 2019	17	49	28	49	29.8%	0.40 [0.18, 0.90]
Zhang 2020	19	84	31	84	38.7%	0.50 [0.25, 0.98]
Subtotal (95% CI)		**216**		**211**	**100.0%**	**0.58 [0.35, 0.98]**
Total events	103		122			
Heterogeneity: Tau² = 0.06; Chi² = 2.83, df = 2 (P = 0.24); I² = 29%						
Test for overall effect: Z = 2.05 (P = 0.04)						
Test for subgroup differences: Chi² = 0.08, df = 1 (P = 0.78), I² = 0%						



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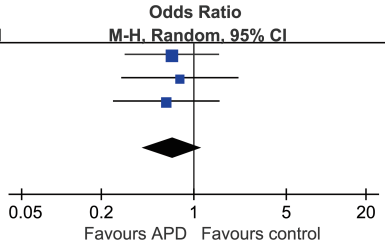
a: microbial infection

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Cohort studies						
Li 2018	43	46	38	42	20.2%	1.51 [0.32, 7.17]
Liu 2015	48	53	44	49	28.8%	1.09 [0.30, 4.02]
Ma 2017	31	43	29	39	51.0%	0.89 [0.33, 2.37]
Subtotal (95% CI)		**142**		**130**	**100.0%**	**1.05 [0.52, 2.12]**
Total events	122		111			
Heterogeneity: Tau² = 0.00; Chi² = 0.32, df = 2 (P = 0.85); I² = 0%						
Test for overall effect: Z = 0.14 (P = 0.89)						
RCTs						
Zhang 2020	78	84	75	84	100.0%	1.56 [0.53, 4.60]
Subtotal (95% CI)		**84**		**84**	**100.0%**	**1.56 [0.53, 4.60]**
Total events	78		75			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.81 (P = 0.42)						
Test for subgroup differences: Chi² = 0.36, df = 1 (P = 0.55), I² = 0%						



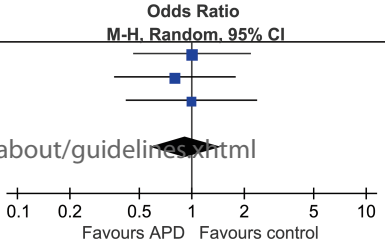
b: sepsis

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Liu 2015	16	53	19	49	41.1%	0.68 [0.30, 1.55]
Liu 2015	7	126	9	129	26.7%	0.78 [0.28, 2.17]
Ma 2017	12	43	15	39	32.2%	0.62 [0.24, 1.57]
Total (95% CI)		**222**		**217**	**100.0%**	**0.69 [0.41, 1.16]**
Total events	35		43			
Heterogeneity: Tau² = 0.00; Chi² = 0.11, df = 2 (P = 0.95); I² = 0%						
Test for overall effect: Z = 1.40 (P = 0.16)						



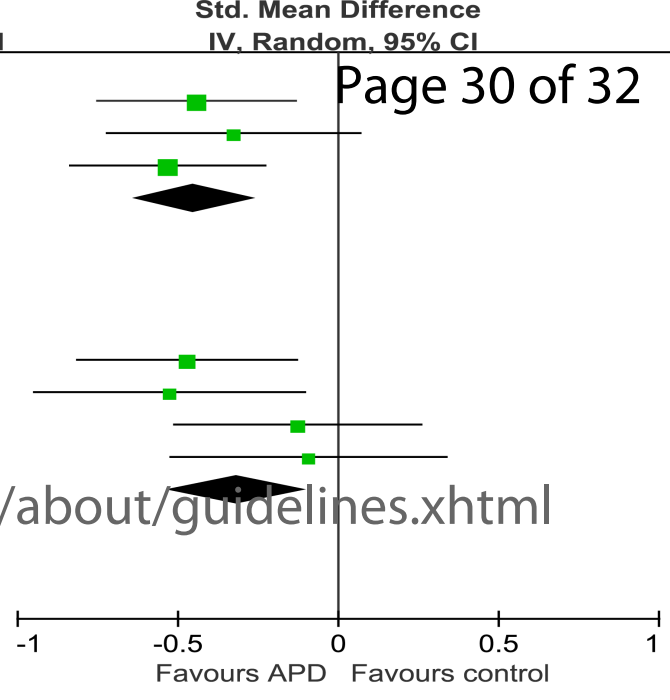
c: bacteremia

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI
Liu 2015	26	53	24	49	36.5%	1.00 [0.46, 2.18]
Liu 2015	12	126	15	129	34.2%	0.80 [0.36, 1.78]
Ma 2017	22	43	20	39	29.3%	1.00 [0.42, 2.37]
Total (95% CI)		**222**		**217**	**100.0%**	**0.93 [0.58, 1.48]**
Total events	60		59			
Heterogeneity: Tau² = 0.00; Chi² = 0.19, df = 2 (P = 0.91); I² = 0%						
Test for overall effect: Z = 0.32 (P = 0.75)						



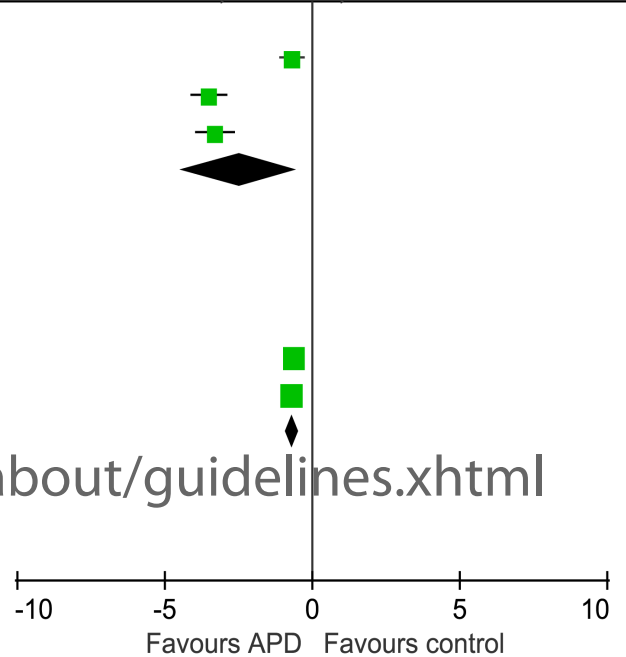
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Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Std. Mean Difference IV, Random, 95% CI
RCTs								
Liang 2016	28.9	10.7	83	35.3	17.5	78	37.7%	-0.44 [-0.76, -0.13]
Luo 2019	23.16	4.98	49	24.65	4.02	49	23.3%	-0.33 [-0.73, 0.07]
Zhang 2020	47.92	9.28	84	52.63	8.31	84	39.0%	-0.53 [-0.84, -0.22]
Subtotal (95% CI)			**216**			**211**	**100.0%**	**-0.45 [-0.64, -0.26]**
Heterogeneity: Tau² = 0.00; Chi² = 0.64, df = 2 (P = 0.72); I² = 0%								
Test for overall effect: Z = 4.59 (P < 0.00001)								
Cohort studies								
Huang 2016	52.78	22.51	68	66.13	33.07	64	30.9%	-0.47 [-0.82, -0.13]
Li 2018	47.93	9.27	46	52.62	8.32	42	22.1%	-0.53 [-0.95, -0.10]
Liu 2015	58.9	31.24	53	63.2	36.15	49	25.7%	-0.13 [-0.52, 0.26]
Ma 2017	55.89	4.59	43	56.28	3.6	39	21.4%	-0.09 [-0.53, 0.34]
Subtotal (95% CI)			**210**			**194**	**100.0%**	**-0.31 [-0.53, -0.10]**
Heterogeneity: Tau² = 0.01; Chi² = 3.64, df = 3 (P = 0.30); I² = 18%								
Test for overall effect: Z = 2.83 (P = 0.005)								
Test for subgroup differences: Chi² = 0.84, df = 1 (P = 0.36), I² = 0%								



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Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Std. Mean Difference IV, Random, 95% CI
Cohort studies								
Li 2018	46,541.93	8,326.67	46	52,518.43	8,874.6	42	33.8%	-0.69 [-1.12, -0.26]
Liu 2015	46,152.2	2,475.13	53	60,013.1	5,041.02	49	33.2%	-3.51 [-4.13, -2.88]
Ma 2017	46,200	2,600	43	60,100	5,400	39	33.0%	-3.30 [-3.98, -2.63]
Subtotal (95% CI)			**142**			**130**	**100.0%**	**-2.49 [-4.46, -0.51]**
Heterogeneity: Tau² = 2.96; Chi² = 71.99, df = 2 (P < 0.00001); I² = 97%								
Test for overall effect: Z = 2.47 (P = 0.01)								
RCTs								
Liang 2016	8,174.1	2,075.8	83	10,572.1	5,041.3	78	49.2%	-0.63 [-0.94, -0.31]
Zhang 2020	46,500	8,400	84	52,600	8,800	84	50.8%	-0.71 [-1.02, -0.39]
Subtotal (95% CI)			**167**			**162**	**100.0%**	**-0.67 [-0.89, -0.44]**
Heterogeneity: Tau² = 0.00; Chi² = 0.12, df = 1 (P = 0.73); I² = 0%								
Test for overall effect: Z = 5.88 (P < 0.00001)								
Test for subgroup differences: Chi² = 3.22, df = 1 (P = 0.07), I² = 68.9%								



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6-7
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Figure 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12-15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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